

# Finite-Strain Hexahedral Finite Elements Truesdell Tangent Representation, Centroidal $\bar{F}$ Hex8, and Sparse Linear Displacement Constraints – Formulation Reference

Continuum formulation, variational formulation, finite element formulation, and solution method

August 2026

## Contents

|           |                                                                            |           |
|-----------|----------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Notation conventions and nomenclature</b>                               | <b>4</b>  |
| <b>2</b>  | <b>Purpose, scope, and organization</b>                                    | <b>6</b>  |
| 2.1       | Normative formulation choices . . . . .                                    | 6         |
| <b>I</b>  | <b>Continuum formulation</b>                                               | <b>6</b>  |
| <b>3</b>  | <b>Configurations, motion, and notation</b>                                | <b>7</b>  |
| 3.1       | Superposed rigid rotations and reported strain measures . . . . .          | 7         |
| 3.2       | Uniaxial-stress stretch and rotation benchmark . . . . .                   | 8         |
| 3.3       | Six-component reporting convention . . . . .                               | 8         |
| 3.4       | Increment and Newton-iteration notation . . . . .                          | 8         |
| <b>4</b>  | <b>Stress measures and the Truesdell rate</b>                              | <b>9</b>  |
| 4.1       | Truesdell rate . . . . .                                                   | 9         |
| <b>5</b>  | <b>The discrete material map and the hierarchy of tangents</b>             | <b>9</b>  |
| <b>6</b>  | <b>Push-forward of the material tangent to a spatial Truesdell tangent</b> | <b>10</b> |
| <b>II</b> | <b>Variational formulation</b>                                             | <b>11</b> |
| <b>7</b>  | <b>How variations are used in this document</b>                            | <b>11</b> |
| <b>8</b>  | <b>Weak form in the current configuration</b>                              | <b>11</b> |

|            |                                                                                 |           |
|------------|---------------------------------------------------------------------------------|-----------|
| <b>9</b>   | <b>Consistent linearization of the internal weak form</b>                       | <b>12</b> |
| 9.1        | Truesdell representation . . . . .                                              | 12        |
| <b>III</b> | <b>Finite element formulation and element technology</b>                        | <b>12</b> |
| <b>10</b>  | <b>Finite element interpolation</b>                                             | <b>13</b> |
| <b>11</b>  | <b>Parent-element integration</b>                                               | <b>13</b> |
| <b>12</b>  | <b>Element residual</b>                                                         | <b>14</b> |
| 12.1       | Voigt convention . . . . .                                                      | 14        |
| <b>13</b>  | <b>Element tangent matrix</b>                                                   | <b>15</b> |
| 13.1       | Truesdell formulation . . . . .                                                 | 15        |
| <b>14</b>  | <b>Supported hexahedral element technologies</b>                                | <b>15</b> |
| <b>15</b>  | <b>Centroidal <math>\bar{F}</math> Hex8 formulation</b>                         | <b>16</b> |
| 15.1       | Projected deformation gradient . . . . .                                        | 16        |
| 15.2       | The material interface is unchanged . . . . .                                   | 17        |
| 15.3       | Modified stress used in the current weak form . . . . .                         | 17        |
| 15.4       | Consistent linearization of the projected material response . . . . .           | 18        |
| 15.5       | Equivalent updated-Lagrangian tangent split . . . . .                           | 18        |
| <b>IV</b>  | <b>Nonlinear solution method</b>                                                | <b>19</b> |
| <b>16</b>  | <b>Pseudo-time discretization and increment control</b>                         | <b>19</b> |
| <b>17</b>  | <b>Gauss-point committed and trial state</b>                                    | <b>20</b> |
| 17.1       | Observation is not a zero-duration constitutive increment . . . . .             | 20        |
| 17.2       | Finite-increment constitutive integration is a separate local problem . . . . . | 21        |
| 17.3       | Geometry and local-failure classification . . . . .                             | 21        |
| <b>18</b>  | <b>Global nonlinear solution with exact linear displacement constraints</b>     | <b>21</b> |
| 18.1       | General affine displacement constraints . . . . .                               | 21        |
| 18.2       | Lagrange-multiplier equilibrium system . . . . .                                | 22        |
| 18.3       | Convergence and residual-based backtracking . . . . .                           | 24        |
| 18.4       | Local error analysis of the search direction . . . . .                          | 25        |
| 18.5       | Indefiniteness, solvability, and physical relevance . . . . .                   | 25        |
| 18.6       | Sparse storage and linear solution strategy . . . . .                           | 26        |

|          |                                                                                        |           |
|----------|----------------------------------------------------------------------------------------|-----------|
| 18.7     | Relation between global Newton and the local material tangent . . . . .                | 26        |
| <b>V</b> | <b>Element and global algorithms</b>                                                   | <b>27</b> |
| 19       | Element algorithm: standard full-integration formulation                               | 27        |
| 20       | Element algorithm: centroidal $\bar{F}$ Hex8                                           | 29        |
| 21       | Global nonlinear solution algorithm                                                    | 31        |
| <b>A</b> | <b>Example constitutive plug-in: finite-strain J2 plasticity</b>                       | <b>32</b> |
| A.1      | Kinematics, elastic response, and history . . . . .                                    | 32        |
| A.2      | Elastic predictor and radial return . . . . .                                          | 32        |
| A.3      | Exact linearization of the discrete update . . . . .                                   | 33        |
| <b>B</b> | <b>Additional tangent remarks and crystal-plasticity differentiation</b>               | <b>34</b> |
| B.1      | Increment notation versus endpoint notation . . . . .                                  | 34        |
| B.2      | Numerical differentiation of a crystal-plasticity update . . . . .                     | 34        |
| <b>C</b> | <b>Reference-form cross-check of the spatial linearization</b>                         | <b>35</b> |
| <b>D</b> | <b>Near-incompressibility and element-technology alternatives</b>                      | <b>35</b> |
| D.1      | How the main remedies alter the present formulation . . . . .                          | 35        |
| D.2      | Published and public implementation references . . . . .                               | 37        |
| <b>E</b> | <b>Hexahedral shape functions and local-node conventions</b>                           | <b>37</b> |
| E.1      | Hex8 . . . . .                                                                         | 37        |
| E.2      | Hex20 . . . . .                                                                        | 37        |
| <b>F</b> | <b>Guide to variations and consistent linearization</b>                                | <b>38</b> |
| F.1      | Variation as a directional derivative . . . . .                                        | 38        |
| F.2      | Product rule and dependency bookkeeping . . . . .                                      | 38        |
| F.3      | Test field versus Newton direction . . . . .                                           | 39        |
| F.4      | Basic kinematic identities . . . . .                                                   | 39        |
| F.5      | Why the spatial test-function gradient varies . . . . .                                | 39        |
| F.6      | Variation of the current volume element . . . . .                                      | 39        |
| F.7      | Example: internal virtual work . . . . .                                               | 39        |
| F.8      | Constitutive variables are varied implicitly through the algorithmic tangent . . . . . | 40        |
| F.9      | Reference-form linearization as a useful check . . . . .                               | 40        |
| <b>G</b> | <b>References</b>                                                                      | <b>40</b> |

# 1 Notation conventions and nomenclature

Continuum-mechanics notation and discrete linear-algebra notation are deliberately separated. This convention is used throughout the document.

| Notation                                                              | Meaning                                                 | Typical examples                                                                                                                    |
|-----------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $J, t, \Delta t$                                                      | scalar                                                  | determinant $J$ , pseudo-time $t$ , increment size $\Delta t$                                                                       |
| $\mathbf{X}, \mathbf{u}, \mathbf{F}, \mathbf{P}, \boldsymbol{\sigma}$ | continuum vector or second-order tensor                 | position/displacement fields, deformation gradient, stress                                                                          |
| $\mathbb{A}, \mathbb{C}$                                              | fourth-order tensor                                     | material and spatial algorithmic tangents                                                                                           |
| $A_{iIjJ}, c_{ijkl}$                                                  | components of fourth-order tensors                      | index form of the preceding tangents                                                                                                |
| $\mathbf{u}, \mathbf{f}, \mathbf{r}$                                  | finite-dimensional algebraic vectors                    | nodal displacement, force, residual                                                                                                 |
| $\mathbf{K}, \mathbf{B}, \mathbf{D}, \mathbf{C}$                      | finite-dimensional matrices                             | stiffness, strain–displacement matrix, Voigt constitutive matrix, constraint matrix                                                 |
| $A, B$<br>$(\cdot)^\top$                                              | finite-element node labels<br>transpose operation       | node $A$ or $B$ in one element distinguished from the upright superscript T, which labels the Truesdell rate/tangent representation |
| alg, int, ext, V, e, p                                                | descriptive labels rather than tensor-component indices | algorithmic, internal, external, Voigt, elastic, and plastic labels                                                                 |
| $g, c$                                                                | integration/projection-point labels                     | Gauss point $g$ , element centroid $c$                                                                                              |
| $i, j, k, l$                                                          | spatial component indices                               | 1, 2, 3                                                                                                                             |
| $I, J, K, L$                                                          | reference/material component indices                    | 1, 2, 3                                                                                                                             |
| $n, n + 1$                                                            | committed increment endpoints                           | accepted states at pseudo-times $t_n, t_{n+1}$                                                                                      |
| $(i)$                                                                 | global Newton iteration                                 | current trial state at $t_{n+1}$                                                                                                    |

Important symbols are summarized below. Symbols are also defined at first substantive use.

| Symbol                                               | Meaning                                                                                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------|
| $\Omega_0$                                           | stress-free reference configuration                                                        |
| $\Omega_t$                                           | generic current configuration at pseudo-time/load parameter $t$                            |
| $\Omega_n, \Omega_{n+1}^{(i)}$                       | committed and current Newton-trial configurations                                          |
| $\mathbf{X}, \mathbf{x}, \mathbf{u}$                 | material position, spatial position, displacement field                                    |
| $\boldsymbol{\varphi}$                               | motion map, $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$                             |
| $\boldsymbol{\xi}$                                   | parent-element coordinate vector $(\xi, \eta, \zeta)^\top$                                 |
| $\mathbf{F}, J$                                      | deformation gradient and $J = \det \mathbf{F}$                                             |
| $\mathbf{L}$                                         | spatial velocity gradient, $\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1}$                  |
| $\mathbf{d}, \mathbf{w}$                             | symmetric rate-of-deformation tensor and skew spin, $\mathbf{L} = \mathbf{d} + \mathbf{w}$ |
| $\boldsymbol{\sigma}, \boldsymbol{\tau}, \mathbf{P}$ | Cauchy, Kirchhoff, and first Piola–Kirchhoff stress                                        |
| $\overset{\text{T}}{\boldsymbol{\sigma}}$            | Truesdell objective rate of Cauchy stress                                                  |

|                                                    |                                                                                                                                                                                                     |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{H}$                                      | committed constitutive history/state variables owned by the material model                                                                                                                          |
| $\theta$                                           | immutable material properties shared by a named material definition, such as elastic moduli                                                                                                         |
| $\rho_g$                                           | immutable material-point data that may vary spatially, such as an initial crystal orientation                                                                                                       |
| $\widehat{\mathcal{H}}_{n+1}^{(i)}$                | provisional material state returned for Newton trial $i$ at the target $t_{n+1}$ ; it is not committed until global convergence                                                                     |
| $\widehat{\mathbf{P}}$                             | endpoint first Piola–Kirchhoff stress returned by the complete discrete material update; its arguments are defined in Eq. (23)                                                                      |
| $\mathbb{A}^{\text{alg}}$                          | algorithmic material tangent $D\mathbf{P}_{n+1}/D\mathbf{F}_{n+1}$ of the complete discrete update, with committed data fixed                                                                       |
| $\mathbb{C}^{\text{T,alg}}$                        | spatial algorithmic tangent in the Truesdell representation                                                                                                                                         |
| $\mathbb{a}^{\text{pf}}$                           | direct push-forward of $\mathbb{A}^{\text{alg}}$ before subtracting stress-dependent terms required by the Truesdell representation                                                                 |
| $\mathbf{h}$                                       | spatial perturbation gradient $\delta\mathbf{F}\mathbf{F}^{-1} = \nabla_x\Delta\mathbf{u}$ in the Newton linearization                                                                              |
| $\mathbf{G}_v$                                     | spatial gradient $\nabla_x\mathbf{v}$ of the virtual-displacement/test field during weak-form linearization                                                                                         |
| $\mathbf{d}_h$                                     | symmetric part $\text{sym}\mathbf{h}$ of the Newton perturbation gradient; it is not a separately integrated physical rate                                                                          |
| $\mathbf{v}$                                       | admissible continuum virtual displacement/test field                                                                                                                                                |
| $\mathbf{v}$                                       | nodal coefficient vector representing $\mathbf{v}^h$ after FE interpolation                                                                                                                         |
| $\Delta\mathbf{u}$                                 | continuum Newton search direction/perturbation used to linearize equilibrium                                                                                                                        |
| $\Delta\mathbf{u}$                                 | nodal coefficient vector of the Newton correction                                                                                                                                                   |
| $\mathbf{s}, \mathbf{d}_V, \mathbf{D}$             | Voigt stress vector, engineering-shear representation of a symmetric spatial tensor, and corresponding $6 \times 6$ tangent matrix                                                                  |
| $\mathbf{f}^{\text{int}}, \mathbf{f}^{\text{ext}}$ | discrete internal and external force vectors                                                                                                                                                        |
| $\mathbf{b}_0$                                     | prescribed dead body force per reference volume, when present                                                                                                                                       |
| $\mathbf{K}$                                       | discrete consistent tangent/Jacobian                                                                                                                                                                |
| $G, G^{\text{int}}, G^{\text{ext}}$                | weak residual and its internal/external virtual-work contributions                                                                                                                                  |
| $\mathbf{C}\mathbf{u} = \mathbf{d}$                | general affine linear displacement constraints                                                                                                                                                      |
| $\lambda$                                          | Lagrange multipliers associated with the constraint equations                                                                                                                                       |
| $\omega_g^x$                                       | current quadrature-volume weight $\det(\mathbf{J}_{x,g})W_g$ at Gauss point $g$                                                                                                                     |
| $N_A, W_g$                                         | shape function associated with node $A$ and parent-space Gauss weight at point $g$                                                                                                                  |
| $\mathbf{J}_0, \mathbf{J}_x$                       | reference and current parent-to-physical Jacobian matrices; in element evaluation $\mathbf{J}_x$ is formed on the current Newton-trial configuration unless stated otherwise                        |
| $D_Q(\cdot), \delta(\cdot)$                        | total/directional derivative with respect to the indicated argument $Q$ , and the corresponding variation; committed quantities named after the semicolon in a discrete material map are held fixed |
| $\mathbf{R}(\mathbf{u})$                           | unconstrained finite-dimensional mechanical residual function $\mathbf{f}^{\text{int}}(\mathbf{u}) - \mathbf{f}^{\text{ext}}$                                                                       |
| $\mathbf{r}_u, \mathbf{r}_c$                       | displacement-equilibrium and constraint residual vectors in one KKT/Newton iteration                                                                                                                |
| $\mathbf{Z}, \mathbf{K}_{\text{red}}$              | basis of $\ker \mathbf{C}$ and reduced constrained tangent $\mathbf{Z}^\top \mathbf{K} \mathbf{Z}$ used in the local solvability discussion                                                         |
| $\alpha_g, q_g$                                    | scalar volumetric-projection factor and its directional logarithmic variation for centroidal $\bar{F}$ Hex8                                                                                         |

|                              |                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| $\bar{F}$                    | centroidally projected deformation gradient used only by Hex8- $\bar{F}$                                                                         |
| $\mathbf{F}_e, \mathbf{F}_p$ | elastic and plastic deformation-gradient factors in a multiplicative finite-strain inelastic decomposition, when that constitutive model is used |

---

## 2 Purpose, scope, and organization

This document is the mathematical formulation reference for a finite-strain, quasi-static finite element solver. It develops the continuum kinematics, Truesdell tangent representation, current-configuration weak form and consistent linearization, standard Hex8 and Hex20 elements, centroidal  $\bar{F}$  Hex8, and the sparse Lagrange-multiplier treatment of general linear displacement constraints. A separate implementation specification translates the final equations and algorithms into software contracts and executable requirements.

The mathematical path is deliberately singular: Cauchy stress is used in the current configuration, the Truesdell rate represents the spatial algorithmic tangent, and the constitutive model is exposed to the element through a discrete finite-deformation update based on  $\mathbf{F}_n$  and  $\mathbf{F}_{n+1}^{(i)}$ . Derivations and explanatory material belong here; code-level array layouts, file formats, and acceptance decks belong in the implementation specification.

### 2.1 Normative formulation choices

- The notation follows Belytschko, Liu, Moran, and Elkhodary, *Nonlinear Finite Elements for Continua and Structures*, second edition, where practical.
- Equilibrium is written in the current configuration and discretized in updated-Lagrangian form.  $\Omega_0$  remains available for total deformation gradients.
- The problem is quasi-static and permits arbitrary finite strain and rotation. Displacement is the only primary FE field.
- The supported solid elements are standard Hex8, centroidal  $\bar{F}$  Hex8, and standard Hex20 serendipity. Hex8 uses  $2^3$  full integration; Hex20 uses  $3^3$  full integration. The  $\bar{F}$  centroid is a projection point, not a constitutive integration point.
- The constitutive tangent is  $D\mathbf{P}_{n+1}/D\mathbf{F}_{n+1}$  for the complete discrete material update. The element formulation owns push-forward, geometric, and  $\bar{F}$  projection terms.
- General affine displacement constraints are imposed with Lagrange multipliers through a sparse KKT system; no dependent displacement DOF elimination is assumed.
- Deformation gradients are derived kinematic quantities reconstructed from nodal positions/displacements. They are not independently advanced Gauss-point state variables.

## Part I

# Continuum formulation

### 3 Configurations, motion, and notation

Let  $\Omega_0$  denote the stress-free reference configuration, with material coordinates  $\mathbf{X}$ . At pseudo-time  $t$  the motion is

$$\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t), \quad \mathbf{u}(\mathbf{X}, t) = \mathbf{x} - \mathbf{X}. \quad (1)$$

The deformation gradient and its Jacobian are

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \mathbf{I} + \frac{\partial \mathbf{u}}{\partial \mathbf{X}}, \quad J = \det \mathbf{F} > 0. \quad (2)$$

Let  $\mathbf{v}_{\text{vel}} = \dot{\mathbf{x}}$  denote the derivative of the motion with respect to  $t$ . When  $t$  is physical time this is the spatial velocity; for a rate-independent pseudo-time parametrization it is a path derivative used to express the same kinematic rate identities. The velocity gradient is

$$\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1}, \quad (3)$$

with symmetric and skew parts

$$\mathbf{d} = \frac{1}{2}(\mathbf{L} + \mathbf{L}^\top), \quad \mathbf{w} = \frac{1}{2}(\mathbf{L} - \mathbf{L}^\top), \quad \mathbf{L} = \mathbf{d} + \mathbf{w}. \quad (4)$$

True physical time is not required by the quasi-static equilibrium equations. The parameter  $t$  may therefore be a monotonically increasing pseudo-time or load parameter for a rate-independent material. If a constitutive model is genuinely rate dependent,  $t$  must instead be assigned its physical-time meaning and the physical  $\Delta t$  must be supplied to the local update.

#### 3.1 Superposed rigid rotations and reported strain measures

Consider a proper orthogonal tensor  $\mathbf{Q}$ , with  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$  and  $\det \mathbf{Q} = 1$ , superposed on a deformation. The rotated motion has

$$\mathbf{F}^+ = \mathbf{Q}\mathbf{F}, \quad J^+ = J. \quad (5)$$

Objectivity of a hyperelastic response requires

$$W(\mathbf{Q}\mathbf{F}) = W(\mathbf{F}), \quad \mathbf{P}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\mathbf{P}(\mathbf{F}), \quad \boldsymbol{\sigma}(\mathbf{Q}\mathbf{F}) = \mathbf{Q}\boldsymbol{\sigma}(\mathbf{F})\mathbf{Q}^\top. \quad (6)$$

This endpoint covariance is the relevant large-rotation verification for the hyperelastic prototype. It tests the complete kinematic, constitutive, stress-transformation, and output path; it should not be described as a numerical integration test of a hypoelastic stress-rate equation.

The Green–Lagrange and Euler–Almansi strains are

$$\mathbf{E} = \frac{1}{2}(\mathbf{F}^\top \mathbf{F} - \mathbf{I}), \quad \mathbf{e} = \frac{1}{2}(\mathbf{I} - \mathbf{F}^{-\top} \mathbf{F}^{-1}). \quad (7)$$

They intentionally report different frames. Under the same superposed rotation,

$$\mathbf{E}^+ = \mathbf{E}, \quad \mathbf{e}^+ = \mathbf{Q}\mathbf{e}\mathbf{Q}^\top. \quad (8)$$

Thus a bar stretched along the reference  $X_1$  direction and then rigidly rotated by  $90^\circ$  in the  $x_1$ – $x_2$  plane retains its dominant  $E_{11}$  component, whereas the dominant spatial stress and Euler–Almansi strain components move from 11 to 22. Reporting both strains makes this frame distinction explicit rather than labeling an unspecified tensor simply as “strain.”

### 3.2 Uniaxial-stress stretch and rotation benchmark

A prescribed axial stretch with fixed lateral stretches is not a uniaxial-stress test. For the compressible neo-Hookean energy used here, the Cauchy stress is

$$\boldsymbol{\sigma} = \frac{1}{J} \left[ \mu(\mathbf{F}\mathbf{F}^\top - \mathbf{I}) + \kappa \ln J \mathbf{I} \right]. \quad (9)$$

Consider  $\mathbf{F}_s = \text{diag}(\lambda, a, a)$ , where  $a > 0$  is the transverse stretch. Vanishing lateral traction requires

$$\mu(a^2 - 1) + \kappa \ln(\lambda a^2) = 0. \quad (10)$$

For positive  $\mu$  and  $\kappa$ , its left-hand side is strictly increasing in  $a$ , ranges from  $-\infty$  to  $+\infty$ , and therefore has exactly one positive root. For tensile  $\lambda > 1$ ,  $\lambda^{-1/2} < a < 1$ : contraction is neither suppressed nor assumed exactly isochoric. The resulting stress is

$$\boldsymbol{\sigma}_s = \sigma_a \mathbf{e}_1 \otimes \mathbf{e}_1, \quad \sigma_a = \frac{\mu(\lambda^2 - a^2)}{\lambda a^2}. \quad (11)$$

After a superposed rotation  $\mathbf{F} = \mathbf{R}_z(\theta) \mathbf{F}_s$ ,

$$\boldsymbol{\sigma} = \sigma_a \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta & 0 \\ \cos \theta \sin \theta & \sin^2 \theta & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (12)$$

Thus  $\sigma_{11} = 0$  and  $\sigma_{22} = \sigma_a$  at  $90^\circ$ . This homogeneous solution can be driven by prescribing its motion on the two end faces, including the compatible transverse contraction, while leaving the lateral surfaces traction-free. Interior and lateral-face nodes not on the ends are equilibrium unknowns. This avoids boundary layers due to rigid transverse clamps. Lateral strains remain nonzero despite zero lateral stress; their values rotate for the spatial strain but not for Green–Lagrange strain.

### 3.3 Six-component reporting convention

For any reported symmetric tensor  $\mathbf{S}$ , use the unscaled representation

$$\mathbf{s}_{\text{out}} = [S_{11} \ S_{22} \ S_{33} \ S_{12} \ S_{23} \ S_{13}]^\top. \quad (13)$$

In particular, strain shear components are tensor entries, not engineering shear strains. This reporting convention is separate from the engineering-shear vectors used in element assembly. For symmetric tensors  $\mathbf{S}$  and  $\mathbf{T}$ , the double contraction is

$$\mathbf{S} : \mathbf{T} = \mathbf{s}_{\text{out}}^\top \text{diag}(1, 1, 1, 2, 2, 2) \mathbf{t}_{\text{out}}. \quad (14)$$

Work and energy calculations must also use mutually conjugate stress and strain measures; the shear weights alone do not make Cauchy stress conjugate to Green–Lagrange strain.

### 3.4 Increment and Newton-iteration notation

The state at  $t_n$  is converged and committed. The next pseudo-time is

$$t_{n+1} = t_n + \Delta t. \quad (15)$$

Within increment  $n \rightarrow n + 1$ , superscript  $(i)$  denotes a global Newton trial:

$$\mathbf{u}_{n+1}^{(i)}, \quad \mathbf{F}_{n+1}^{(i)}, \quad \boldsymbol{\sigma}_{n+1}^{(i)}, \quad \widehat{\mathcal{H}}_{n+1}^{(i)}. \quad (16)$$

Every local constitutive evaluation at iteration  $i$  starts from the same committed data

$$\{\mathbf{F}_n, \mathcal{H}_n\} \quad (17)$$

and integrates to the current trial endpoint  $\mathbf{F}_{n+1}^{(i)}$ .



## 4 Stress measures and the Truesdell rate

The Cauchy stress is  $\boldsymbol{\sigma}$ . The Kirchhoff and first Piola-Kirchhoff stresses are

$$\boldsymbol{\tau} = J\boldsymbol{\sigma}, \quad \mathbf{P} = J\boldsymbol{\sigma}\mathbf{F}^{-\top}. \quad (18)$$

Balance of angular momentum implies  $\boldsymbol{\sigma} = \boldsymbol{\sigma}^\top$  for the non-polar continua considered here.

### 4.1 Truesdell rate

The Truesdell rate of Cauchy stress is

$$\dot{\boldsymbol{\sigma}}^\top = \dot{\boldsymbol{\sigma}} - \mathbf{L}\boldsymbol{\sigma} - \boldsymbol{\sigma}\mathbf{L}^\top + \text{tr}(\mathbf{L})\boldsymbol{\sigma}. \quad (19)$$

Since  $\boldsymbol{\tau} = J\boldsymbol{\sigma}$  and  $\dot{J} = J\text{tr}(\mathbf{L})$ ,

$$\mathcal{L}_{\mathbf{v}_{\text{vel}}}\boldsymbol{\tau} = \dot{\boldsymbol{\tau}} - \mathbf{L}\boldsymbol{\tau} - \boldsymbol{\tau}\mathbf{L}^\top = J\dot{\boldsymbol{\sigma}}^\top. \quad (20)$$

Thus the Truesdell rate of Cauchy stress is the Lie derivative of Kirchhoff stress divided by  $J$ .

A spatial constitutive relation written using this rate has the form

$$\dot{\boldsymbol{\sigma}}^\top = \mathbb{c}^\top : \mathbf{d}. \quad (21)$$

For an incrementally integrated material, the corresponding  $\mathbb{c}^{\text{T,alg}}$  used in the finite element tangent is understood as an *algorithmic spatial tangent*, not necessarily the instantaneous continuum modulus.

## 5 The discrete material map and the hierarchy of tangents

The data entering a constitutive law separate naturally into three categories. The vector or structured collection  $\boldsymbol{\theta}$  contains fixed properties shared by a material definition. The data  $\boldsymbol{\rho}_g$  are also fixed during the analysis but may differ between material points; an initial crystal orientation is a representative example. Only  $\mathcal{H}_g$  contains evolving history. This distinction prevents spatial labels from being treated as constitutive history and permits several property sets to share the same constitutive equations.

The local material update is considered as a discrete algorithm

$$\mathcal{M} : \left( \mathbf{F}_n, \mathbf{F}_{n+1}^{(i)}, \mathcal{H}_n; \boldsymbol{\theta}, \boldsymbol{\rho}_g, t_n, t_{n+1} \right) \longrightarrow \left( \mathbf{P}_{n+1}^{(i)}, \mathbb{A}_{n+1}^{\text{alg},(i)}, \widehat{\mathcal{H}}_{n+1}^{(i)} \right). \quad (22)$$

Write the stress component of this discrete update as

$$\mathbf{P}_{n+1} = \widehat{\mathbf{P}}(\mathbf{F}_{n+1}; \mathbf{F}_n, \mathcal{H}_n, \boldsymbol{\theta}, \boldsymbol{\rho}_g, \Delta t). \quad (23)$$

The hat does not denote a new stress measure. It denotes the algorithmically returned endpoint stress after the complete local integration from the committed state  $(\mathbf{F}_n, \mathcal{H}_n)$  to the specified endpoint  $\mathbf{F}_{n+1}$ . The material properties and point data parameterize that map but are not advanced by it. With all arguments after the semicolon held fixed,

$$\boxed{\mathbb{A}_{n+1}^{\text{alg}} = D_{\mathbf{F}_{n+1}} \widehat{\mathbf{P}}(\mathbf{F}_{n+1}; \mathbf{F}_n, \mathcal{H}_n, \boldsymbol{\theta}, \boldsymbol{\rho}_g, \Delta t)} \quad (24)$$

is the derivative of the complete discrete update. It contains the response of all locally integrated internal variables.

Three distinct tangents should not be conflated:

$$\mathbb{A}^{\text{frozen}} = \left. \frac{\partial \mathbf{P}}{\partial \mathbf{F}} \right|_{\mathcal{H}}, \quad (25)$$

$$\mathbb{A}^{\text{alg}} = \frac{D\mathbf{P}_{n+1}}{D\mathbf{F}_{n+1}}, \quad (26)$$

$$\mathbf{K}_e = \frac{\partial \mathbf{f}^{e,\text{int}}}{\partial \mathbf{u}_e}. \quad (27)$$

The first is a frozen-state material derivative, the second is a local algorithmic material tangent, and the third is the element contribution to the global Newton Jacobian. The finite element linearization maps the second into the third and adds the geometric contribution.

## 6 Push-forward of the material tangent to a spatial Truesdell tangent

The relation

$$\mathbf{P} = J\boldsymbol{\sigma}\mathbf{F}^{-\top} \quad (28)$$

provides the bridge between the neutral material tangent and the spatial tangent required by the current-configuration weak form.

Let an infinitesimal perturbation of the endpoint configuration be  $\delta\mathbf{F}$  and define the spatial perturbation gradient

$$\mathbf{h} = \delta\mathbf{F}\mathbf{F}^{-1}, \quad \delta\mathbf{F} = \mathbf{h}\mathbf{F}. \quad (29)$$

The variation of Eq. (18) is

$$\delta\mathbf{P} = J \left[ \delta\boldsymbol{\sigma} + \text{tr}(\mathbf{h})\boldsymbol{\sigma} - \boldsymbol{\sigma}\mathbf{h}^\top \right] \mathbf{F}^{-\top}. \quad (30)$$

Multiplying by  $\mathbf{F}^\top/J$  gives

$$\frac{1}{J}\delta\mathbf{P}\mathbf{F}^\top = \delta\boldsymbol{\sigma} + \text{tr}(\mathbf{h})\boldsymbol{\sigma} - \boldsymbol{\sigma}\mathbf{h}^\top. \quad (31)$$

Define the Truesdell-type endpoint variation

$$\delta^{\text{T}}\boldsymbol{\sigma} = \delta\boldsymbol{\sigma} - \mathbf{h}\boldsymbol{\sigma} - \boldsymbol{\sigma}\mathbf{h}^\top + \text{tr}(\mathbf{h})\boldsymbol{\sigma}. \quad (32)$$

Then

$$\boxed{\delta^{\text{T}}\boldsymbol{\sigma} = \frac{1}{J}\delta\mathbf{P}\mathbf{F}^\top - \mathbf{h}\boldsymbol{\sigma}}. \quad (33)$$

Using  $\delta\mathbf{P} = \mathbb{A}^{\text{alg}} : \delta\mathbf{F}$  and Eq. (29), introduce the fourth-order push-forward tensor  $\mathbf{a}^{\text{pf}}$  through its components

$$\boxed{a_{ij\,km}^{\text{pf}} = \frac{1}{J}A_{iI\,kK}^{\text{alg}}F_{jI}F_{mK}} \quad (34)$$

Equation (33) becomes

$$\delta^{\text{T}}\sigma_{ij} = \left( a_{ij\,km}^{\text{pf}} - \delta_{ik}\sigma_{mj} \right) h_{km}. \quad (35)$$

For an objective material algorithm, the constitutive part depends on the symmetric perturbation

$$\mathbf{d}_h = \text{sym } \mathbf{h}. \quad (36)$$

A convenient minor-symmetric spatial algorithmic tangent is therefore

$$\boxed{c_{ij\,km}^{\text{T,alg}} = \frac{1}{2} \left[ a_{ij\,km}^{\text{pf}} + a_{ij\,mk}^{\text{pf}} - \delta_{ik}\sigma_{mj} - \delta_{im}\sigma_{kj} \right]} \quad (37)$$

so that

$$\delta^T \boldsymbol{\sigma} = \mathbb{c}^{\text{T,alg}} : \mathbf{d}_h. \quad (38)$$

For an exact objective constitutive update, the appropriate output symmetries follow analytically. In a numerical implementation, loss of expected symmetries is a useful diagnostic and should not automatically be hidden by symmetrization unless the intended mathematical symmetry has first been established.

## Part II

# Variational formulation

## 7 How variations are used in this document

Two independent directions appear in the variational formulation and should not be conflated. The field  $\mathbf{v}$  is an admissible virtual displacement used to test equilibrium. The field  $\Delta \mathbf{u}$  is the Newton search direction used to linearize the nonlinear residual about the current trial configuration. During the Newton linearization, the nodal coefficients of  $\mathbf{v}$  are held fixed.

A variation acts only on quantities that depend on the perturbed unknown. For a product  $A(\mathbf{u})B(\mathbf{u})$ ,

$$\delta(AB) = (\delta A)B + A(\delta B), \quad (39)$$

but if  $B$  is independent of  $\mathbf{u}$ , then  $\delta B = 0$  and only the first term remains. The apparent asymmetry that often occurs in nonlinear finite-element derivations therefore reflects dependency, not a suspension of the product rule.

Updated-Lagrangian expressions contain an important example. The test-function coefficients are fixed, but its spatial gradient depends on the current coordinate map. Consequently

$$\delta \mathbf{v} = \mathbf{0} \quad \text{in coefficient space, while} \quad \delta(\nabla_x \mathbf{v}) \neq \mathbf{0}. \quad (40)$$

In contrast,  $\text{Grad}_X \mathbf{v}$  is unchanged when the same test field is expressed with derivatives with respect to the fixed reference coordinates. Likewise, the current volume element  $dv$  varies, whereas the reference volume element  $dV$  does not.

Appendix F develops these rules in more detail and works through the variations used in the internal-force linearization, including  $\delta J$ ,  $\delta \mathbf{F}^{-\top}$ ,  $\delta(\nabla_x \mathbf{v})$ , and the distinction between explicit and constitutively condensed dependencies.

## 8 Weak form in the current configuration

Let  $\mathbf{v}$  denote an admissible virtual displacement. The quasi-static weak equilibrium statement in the current configuration  $\Omega_t$  is

$$G(\mathbf{u}, \mathbf{v}) = G^{\text{int}}(\mathbf{u}, \mathbf{v}) - G^{\text{ext}}(\mathbf{v}) = 0, \quad (41)$$

where

$$G^{\text{int}} = \int_{\Omega_t} \boldsymbol{\sigma} : \nabla_x \mathbf{v} \, dv = \int_{\Omega_t} \boldsymbol{\sigma} : \text{sym}(\nabla_x \mathbf{v}) \, dv \quad (42)$$

because  $\boldsymbol{\sigma}$  is symmetric.

After spatial discretization the continuum test field is represented by nodal coefficients,

$$\mathbf{v}^h(\mathbf{X}) = \sum_A N_A(\mathbf{X}) \mathbf{v}_A, \quad \mathbf{v} = [\mathbf{v}_1^\top \cdots \mathbf{v}_N^\top]^\top. \quad (43)$$

Thus  $\mathbf{v}$  denotes the continuum test field, while  $\mathbf{v}$  denotes its finite-dimensional nodal coefficient vector.

The present formulation admits dead external loading only. A convenient representation is a prescribed body force per reference volume  $\mathbf{b}_0$  together with dead nodal forces,

$$G^{\text{ext}} = \int_{\Omega_0} \mathbf{b}_0 \cdot \mathbf{v} \, dV + (\mathbf{v})^\top \mathbf{f}_{\text{nodal}}^{\text{ext}}. \quad (44)$$

These loads are independent of the current unknown configuration, so their tangent vanishes. Prescribed tractions per current area, pressure loads, and other follower loads are excluded because they would require an external-load linearization.

## 9 Consistent linearization of the internal weak form

Let  $\Delta \mathbf{u}$  be a Newton correction and define its spatial gradient

$$\mathbf{h} = \nabla_x \Delta \mathbf{u}, \quad \mathbf{d}_h = \text{sym } \mathbf{h}. \quad (45)$$

The directional derivative of  $G^{\text{int}}$  is evaluated at the current Newton trial configuration  $\Omega_{n+1}^{(i)}$ ; for readability the subscript is suppressed in the following local linearization.

Define

$$\mathbf{G}_v = \nabla_x \mathbf{v}. \quad (46)$$

Under an infinitesimal perturbation of that trial configuration,

$$\delta(\text{dv}) = \text{tr}(\mathbf{h}) \, \text{dv}, \quad (47)$$

$$\delta \mathbf{G}_v = -\mathbf{G}_v \mathbf{h}. \quad (48)$$

Therefore

$$DG^{\text{int}}[\Delta \mathbf{u}] = \int_{\Omega_t} [\mathbf{G}_v : \delta \boldsymbol{\sigma} - \boldsymbol{\sigma} : (\mathbf{G}_v \mathbf{h}) + \text{tr}(\mathbf{h}) \boldsymbol{\sigma} : \mathbf{G}_v] \, \text{dv}. \quad (49)$$

### 9.1 Truesdell representation

From the Truesdell constitutive variation,

$$\delta \boldsymbol{\sigma} = \mathbb{C}^{\text{T,alg}} : \mathbf{d}_h + \mathbf{h} \boldsymbol{\sigma} + \boldsymbol{\sigma} \mathbf{h}^\top - \text{tr}(\mathbf{h}) \boldsymbol{\sigma}. \quad (50)$$

Insert Eq. (50) into Eq. (49). The volume-change terms cancel. The term  $\mathbf{G}_v : \boldsymbol{\sigma} \mathbf{h}^\top$  cancels  $\boldsymbol{\sigma} : (\mathbf{G}_v \mathbf{h})$ . The remaining result is

$$DG^{\text{int}}[\Delta \mathbf{u}] = \int_{\Omega_t} \text{sym}(\mathbf{G}_v) : \mathbb{C}^{\text{T,alg}} : \mathbf{d}_h \, \text{dv} + \int_{\Omega_t} (G_v)_{ij} h_{ik} \sigma_{kj} \, \text{dv}. \quad (51)$$

The first term gives the material tangent contribution. The second is the standard geometric or initial-stress contribution.

This result is particularly useful because the geometric stiffness emerges from the linearization rather than being appended as an independent finite element technology.

## Part III

# Finite element formulation and element technology

## 10 Finite element interpolation

For an element with  $n_e$  nodes,

$$\mathbf{x}^h(\boldsymbol{\xi}) = \sum_{A=1}^{n_e} N_A(\boldsymbol{\xi}) \mathbf{x}_A, \quad \mathbf{u}^h(\boldsymbol{\xi}) = \sum_{A=1}^{n_e} N_A(\boldsymbol{\xi}) \mathbf{u}_A, \quad (52)$$

where  $\boldsymbol{\xi} = (\xi, \eta, \zeta)^\top \in [-1, 1]^3$  are parent coordinates.

The reference and current element maps are

$$\mathbf{X}(\boldsymbol{\xi}) = \sum_A N_A(\boldsymbol{\xi}) \mathbf{X}_A, \quad (53)$$

$$\mathbf{x}(\boldsymbol{\xi}) = \sum_A N_A(\boldsymbol{\xi}) \mathbf{x}_A. \quad (54)$$

Define the Jacobian matrices

$$\mathbf{J}_0 = \frac{\partial \mathbf{X}}{\partial \boldsymbol{\xi}}, \quad \mathbf{J}_x = \frac{\partial \mathbf{x}}{\partial \boldsymbol{\xi}}. \quad (55)$$

With column-gradient convention,

$$\nabla_X N_A = \mathbf{J}_0^{-\top} \nabla_{\boldsymbol{\xi}} N_A, \quad (56)$$

$$\nabla_x N_A = \mathbf{J}_x^{-\top} \nabla_{\boldsymbol{\xi}} N_A. \quad (57)$$

The total deformation gradient at a Gauss point follows either from nodal coordinates or from the two Jacobian maps:

$$\boxed{\mathbf{F} = \sum_A \mathbf{x}_A \otimes \nabla_X N_A = \mathbf{J}_x \mathbf{J}_0^{-1}} \quad (58)$$

for the consistent matrix convention used here.

Current volume integration is

$$dv = \det(\mathbf{J}_x) d\xi d\eta d\zeta. \quad (59)$$

Reference and current Jacobians must remain orientation-consistent, and positive element Jacobians are required at all integration points.

## 11 Parent-element integration

For the standard Hex8 element and the centroidal  $\bar{F}$  Hex8 element, use the two-point Gauss-Legendre rule in each parent coordinate,

$$\xi_g \in \left\{ -\frac{1}{\sqrt{3}}, +\frac{1}{\sqrt{3}} \right\}, \quad w_g = 1, \quad (60)$$

which gives eight integration points. The parent-space centroid

$$\boldsymbol{\xi}_c = (0, 0, 0)^\top \quad (61)$$

is used by the  $\bar{F}$  Hex8 element only for the volumetric projection; no constitutive integration or material history is stored there.

For Hex20, use the three-point Gauss-Legendre rule in each parent coordinate:

$$\xi_g \in \left\{ -\sqrt{\frac{3}{5}}, 0, +\sqrt{\frac{3}{5}} \right\}, \quad w_g \in \left\{ \frac{5}{9}, \frac{8}{9}, \frac{5}{9} \right\}, \quad (62)$$

which gives 27 integration points.

For any of these current-configuration integrals,

$$\int_{\Omega_e} q(\mathbf{x}) \, dv \approx \sum_{g=1}^{n_g} q(\boldsymbol{\xi}_g) \det \mathbf{J}_x(\boldsymbol{\xi}_g) W_g, \quad n_g = \begin{cases} 8, & \text{Hex8,} \\ 27, & \text{Hex20,} \end{cases} \quad (63)$$

where  $W_g = w_{g1} w_{g2} w_{g3}$ . Define the current quadrature-volume weight

$$\boxed{\omega_g^x = \det \mathbf{J}_{x,g} W_g.} \quad (64)$$

Consequently,

$$\mathbf{f}^{e,\text{int}} \approx \sum_g \mathbf{B}_g^\top \mathbf{s}_g \omega_g^x, \quad (65)$$

$$\mathbf{K}_{\text{mat}} \approx \sum_g \mathbf{B}_g^\top \mathbf{D}_g^{\text{T,alg}} \mathbf{B}_g \omega_g^x, \quad (66)$$

$$\mathbf{K}_{\text{geo},AB} \approx \sum_g \left[ (\nabla_x N_A)_g^\top \boldsymbol{\sigma}_g (\nabla_x N_B)_g \right] \mathbf{I}_3 \omega_g^x. \quad (67)$$

## 12 Element residual

Using the nodal coefficients  $\mathbf{v}_A$  of the virtual displacement  $\mathbf{v}^h$ , Eq. (42) gives

$$G^{e,\text{int}} = \sum_A v_{Ai} \int_{\Omega_e} N_{A,j} \sigma_{ij} \, dv. \quad (68)$$

Hence the internal-force contribution of node  $A$  is

$$\boxed{(f^{e,\text{int}})_{Ai} = \int_{\Omega_e} N_{A,j} \sigma_{ij} \, dv} \quad (69)$$

or, in standard matrix form,

$$\boxed{\mathbf{f}^{e,\text{int}} = \int_{\Omega_e} \mathbf{B}^\top \mathbf{s} \, dv.} \quad (70)$$

### 12.1 Voigt convention

Use stress vector

$$\mathbf{s} = [\sigma_{11}, \sigma_{22}, \sigma_{33}, \sigma_{12}, \sigma_{23}, \sigma_{31}]^\top \quad (71)$$

and engineering-strain-like vector

$$\mathbf{d}_V = [d_{11}, d_{22}, d_{33}, 2d_{12}, 2d_{23}, 2d_{31}]^\top. \quad (72)$$

For node  $A$ ,

$$\mathbf{B}_A = \begin{bmatrix} N_{A,1} & 0 & 0 \\ 0 & N_{A,2} & 0 \\ 0 & 0 & N_{A,3} \\ N_{A,2} & N_{A,1} & 0 \\ 0 & N_{A,3} & N_{A,2} \\ N_{A,3} & 0 & N_{A,1} \end{bmatrix}, \quad \mathbf{B} = [\mathbf{B}_1 \cdots \mathbf{B}_{n_e}]. \quad (73)$$

Let the ordered symmetric index pairs be

$$\mathcal{I} = [(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (3, 1)]. \quad (74)$$

For a fourth-order tangent  $c_{ijkl}$  acting on a symmetric tensor  $d_{kl}$ , the  $6 \times 6$  matrix  $\mathbf{D}$  satisfying  $\delta \mathbf{s} = \mathbf{D} \mathbf{d}_V$  is defined literally by

$$D_{IJ} = \begin{cases} c_{ij\,kk}, & \mathcal{I}_J = (k, k), \\ \frac{1}{2}(c_{ij\,kl} + c_{ij\,lk}), & \mathcal{I}_J = (k, l), \, k \neq l, \end{cases} \quad \mathcal{I}_I = (i, j). \quad (75)$$

If the expected minor symmetry  $c_{ijkl} = c_{ijlk}$  has been established, the shear-column entry is simply  $c_{ij\,kl}$ . No additional factor of two is inserted because the factor is already carried by the engineering-shear component  $2d_{kl}$ . The six-row representation presumes the physically required symmetric Cauchy-stress variation; unexpected loss of output-pair symmetry is a diagnostic rather than something to hide by automatic symmetrization. This convention is local to the FE matrix representation; the tensor derivation itself remains independent of Voigt notation.

## 13 Element tangent matrix

### 13.1 Truesdell formulation

Equation (51) produces

$$\boxed{\mathbf{K}_e = \mathbf{K}_{\text{mat}} + \mathbf{K}_{\text{geo}}} \quad (76)$$

with

$$\boxed{\mathbf{K}_{\text{mat}} = \int_{\Omega_e} \mathbf{B}^\top \mathbf{D}^{\text{T,alg}} \mathbf{B} \, dv} \quad (77)$$

and the geometric block between nodes  $A$  and  $B$

$$\boxed{(K_{\text{geo}})_{Ai\,Bk} = \delta_{ik} \int_{\Omega_e} N_{A,j} \sigma_{jl} N_{B,l} \, dv.} \quad (78)$$

Equivalently,

$$\mathbf{K}_{\text{geo},AB} = \left[ \int_{\Omega_e} (\nabla_x N_A)^\top \boldsymbol{\sigma} \nabla_x N_B \, dv \right] \mathbf{I}_3. \quad (79)$$

## 14 Supported hexahedral element technologies

The continuum and variational formulation is common to all three element formulations. They differ only in interpolation, quadrature, and, for one element, the volumetric projection used before the constitutive call.

| Formulation      | Interpolation                    | Quadrature                                           | Constitutive kinematics                                                             |
|------------------|----------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>hex8</b>      | 8-node tri-linear                | $2 \times 2 \times 2$                                | Raw $\mathbf{F}_n^g, \mathbf{F}_{n+1}^g$                                            |
| <b>hex8_fbar</b> | 8-node tri-linear                | $2 \times 2 \times 2$ plus<br>centroid<br>projection | $\bar{\mathbf{F}}_n^g, \bar{\mathbf{F}}_{n+1}^g$ through the same<br>callback slots |
| <b>hex20</b>     | 20-node quadratic<br>serendipity | $3 \times 3 \times 3$                                | Raw $\mathbf{F}_n^g, \mathbf{F}_{n+1}^g$                                            |

The standard full-integration algorithm in Section 19 is therefore one kernel parameterized by shape functions and quadrature and covers **hex8** and **hex20**. Section 20 gives the distinct centroidal  $\bar{\mathbf{F}}$  Hex8 wrapper. All three supported formulations return the same element-level objects: internal force, consistent tangent, and trial material-point state.

## 15 Centroidal $\bar{\mathbf{F}}$ Hex8 formulation

The classical  $\bar{\mathbf{F}}$  construction is used here only for the tri-linear Hex8 element. The purpose is to reduce the number of independent volumetric constraints while leaving the deviatoric part of the deformation gradient local to each Gauss point. The formulation is displacement-only and introduces no pressure degree of freedom.

### 15.1 Projected deformation gradient

At Gauss point  $g$ , let

$$\mathbf{F}_g = \mathbf{J}_{x,g} \mathbf{J}_{0,g}^{-1}, \quad J_g = \det \mathbf{F}_g. \quad (80)$$

At the element centroid  $c$ , evaluate the same isoparametric maps and define

$$\mathbf{F}_c = \mathbf{J}_{x,c} \mathbf{J}_{0,c}^{-1}, \quad J_c = \det \mathbf{F}_c. \quad (81)$$

For three dimensions define

$$\alpha_g = \left( \frac{J_c}{J_g} \right)^{1/3}, \quad \bar{\mathbf{F}}_g = \alpha_g \mathbf{F}_g. \quad (82)$$

Then

$$\det \bar{\mathbf{F}}_g = J_c \quad (83)$$

for every Gauss point. Thus every integration point receives the same element-level volumetric deformation, while

$$J_g^{-1/3} \mathbf{F}_g = J_c^{-1/3} \bar{\mathbf{F}}_g \quad (84)$$

shows that its isochoric deformation is unchanged.

The projection is applied independently at the committed configuration  $\Omega_n$  and at every Newton-trial configuration  $\Omega_{n+1}^{(i)}$  associated with the fixed target  $t_{n+1}$ :

$$\bar{\mathbf{F}}_n^g = \alpha_{n,g} \mathbf{F}_n^g, \quad \bar{\mathbf{F}}_{n+1}^{g,(i)} = \alpha_{n+1,g}^{(i)} \mathbf{F}_{n+1}^{g,(i)}. \quad (85)$$

The centroid has no stress, constitutive state, or return-mapping solve associated with it.



## 15.2 The material interface is unchanged

The constitutive update is deliberately unaware of the element technology. Define the kinematic argument supplied by the element wrapper as

$$\mathbf{F}^{\text{mat},g} = \begin{cases} \mathbf{F}_g, & \text{standard element,} \\ \bar{\mathbf{F}}_g, & \bar{F} \text{ Hex8 element.} \end{cases} \quad (86)$$

Both elements therefore call exactly the same routine,

$$\boxed{\mathcal{M} \left( \mathbf{F}_n^{\text{mat},g}, \mathbf{F}_{n+1}^{\text{mat},g,(i)}, \mathcal{H}_n^g, t_n, t_{n+1}, \dots \right) \rightarrow \left( \mathbf{P}_{n+1}^{\text{mat},g,(i)}, \mathbb{A}_{n+1,g}^{\text{mat,alg},(i)}, \hat{\mathcal{H}}_{n+1}^{g,(i)} \right).} \quad (87)$$

For the  $\bar{F}$  element the returned quantities are denoted  $\bar{\mathbf{P}}_g$  and  $\bar{\mathbb{A}}_g^{\text{alg}}$  below, with

$$\bar{\mathbb{A}}_g^{\text{alg}} = \frac{D\bar{\mathbf{P}}_g}{D\bar{\mathbf{F}}_g} \quad (88)$$

for the complete discrete constitutive update. A crystal-plasticity routine therefore performs the same return mapping and returns the same kinds of quantities as it does for a standard element; only the deformation gradients supplied by the element wrapper differ.

This point is important for history-dependent materials. On every global Newton iteration, the  $\bar{F}$  material update is rerun from the committed pair  $\bar{\mathbf{F}}_n^g, \mathcal{H}_n^g$  to the current trial endpoint  $\bar{\mathbf{F}}_{n+1}^{g,(i)}$ . Raw  $\mathbf{F}_g$  is used by the element geometry, but the constitutive history for this element is driven consistently by  $\bar{\mathbf{F}}_g$ .

## 15.3 Modified stress used in the current weak form

From the material return at  $\bar{\mathbf{F}}_g$ , define the associated Cauchy stress

$$\boxed{\bar{\boldsymbol{\sigma}}_g = \frac{1}{J_c} \bar{\mathbf{P}}_g \bar{\mathbf{F}}_g^\top.} \quad (89)$$

The current-configuration internal virtual work keeps exactly the same spatial form as the standard element,

$$\boxed{\bar{G}^{e,\text{int}} = \int_{\Omega_e} \nabla_x \mathbf{v} : \bar{\boldsymbol{\sigma}} \, dv.} \quad (90)$$

Consequently, the ordinary spatial  $\mathbf{B}$  matrix and the actual current volume  $dv$  are used in the residual. The projection acts through the stress supplied to this otherwise unchanged weak form.

It is useful to pull Eq. (90) back to  $\Omega_0$  for the tangent derivation. Define an effective first Piola stress conjugate to the *raw* deformation gradient:

$$\mathbf{P}_g^{\text{eff}} = J_g \bar{\boldsymbol{\sigma}}_g \mathbf{F}_g^{-\top} \quad (91)$$

$$= \boxed{\alpha_g^{-2} \bar{\mathbf{P}}_g}. \quad (92)$$

Hence

$$\bar{G}^{e,\text{int}} = \int_{\Omega_{0e}} \text{Grad } \mathbf{v} : \mathbf{P}^{\text{eff}} \, dV, \quad (93)$$

and the nodal internal force may be evaluated equivalently as

$$\boxed{\bar{f}_{Ai}^{e,\text{int}} = \sum_g N_{A,I}^g (P_{iI}^{\text{eff}})_g \det \mathbf{J}_{0,g} W_g.} \quad (94)$$

Equation (94) and the spatial residual  $\sum_g \mathbf{B}_g^\top \bar{\mathbf{s}}_g \det \mathbf{J}_{x,g} W_g$  are algebraically identical.

## 15.4 Consistent linearization of the projected material response

Let  $\Delta \mathbf{u}$  be an arbitrary Newton perturbation direction and denote its induced kinematic variations by

$$\delta \mathbf{F}_g = \text{Grad } \Delta \mathbf{u}|_g, \quad \delta \mathbf{F}_c = \text{Grad } \Delta \mathbf{u}|_c. \quad (95)$$

The directional change of the scalar projection factor is

$$q_g := \frac{\delta \alpha_g}{\alpha_g} = \frac{1}{3} \left( \mathbf{F}_c^{-\top} : \delta \mathbf{F}_c - \mathbf{F}_g^{-\top} : \delta \mathbf{F}_g \right). \quad (96)$$

Therefore

$$\delta \bar{\mathbf{F}}_g = \alpha_g (\delta \mathbf{F}_g + q_g \mathbf{F}_g) \quad (97)$$

and, because  $\mathbf{P}^{\text{eff}} = \alpha^{-2} \bar{\mathbf{P}}$ ,

$$\delta \mathbf{P}_g^{\text{eff}} = \alpha_g^{-2} \left[ \bar{\mathbb{A}}_g^{\text{alg}} : \delta \bar{\mathbf{F}}_g - 2q_g \bar{\mathbf{P}}_g \right]. \quad (98)$$

Equations (96)–(98) are the central implementation formulas. They require no new material tangent: the element uses exactly the  $D\mathbf{P}/D\mathbf{F}$  object returned by the constitutive update, evaluated at the deformation gradient that the wrapper supplied.

For explicit matrix assembly, define

$$\mathbf{Q}_g = \alpha_g^{-2} \left[ \bar{\mathbb{A}}_g^{\text{alg}} : \bar{\mathbf{F}}_g - 2\bar{\mathbf{P}}_g \right]. \quad (99)$$

For a unit perturbation of nodal degree of freedom  $(B, k)$ ,

$$\delta \mathbf{F}_g^{Bk} = \mathbf{e}_k \otimes \nabla_X N_B|_g, \quad (100)$$

$$\delta \mathbf{F}_c^{Bk} = \mathbf{e}_k \otimes \nabla_X N_B|_c, \quad (101)$$

so that

$$q_g^{Bk} = \frac{1}{3} \left[ (\mathbf{F}_c^{-\top})_{kK} N_{B,K}^c - (\mathbf{F}_g^{-\top})_{kK} N_{B,K}^g \right]. \quad (102)$$

Substitution into Eq. (98) gives the exact element block

$$\begin{aligned} (\mathbf{K}_e^{\bar{F}})_{Ai, Bk} = \sum_g \det \mathbf{J}_{0,g} W_g N_{A,I}^g \left\{ \alpha_g^{-1} (\bar{\mathbb{A}}_g^{\text{alg}})_{iIkK} N_{B,K}^g \right. \\ \left. + \frac{1}{3} (Q_g)_{iI} \left[ (\mathbf{F}_c^{-\top})_{kK} N_{B,K}^c - (\mathbf{F}_g^{-\top})_{kK} N_{B,K}^g \right] \right\}. \end{aligned} \quad (103)$$

This tangent is generally nonsymmetric, even for a conservative material, because the classical centroidal  $\bar{F}$  projection modifies the trial kinematics without symmetrically modifying the test side of the weak form.

## 15.5 Equivalent updated-Lagrangian tangent split

Equation (103) is convenient for implementation and verification, but the same tangent can be expressed directly in the current Newton-trial configuration. Define

$$\mathbf{h}_g = \delta \mathbf{F}_g \mathbf{F}_g^{-1}, \quad \mathbf{h}_c = \delta \mathbf{F}_c \mathbf{F}_c^{-1}, \quad q_g = \frac{1}{3} (\text{tr } \mathbf{h}_c - \text{tr } \mathbf{h}_g), \quad (104)$$

and

$$\bar{\mathbf{h}}_g = \mathbf{h}_g + q_g \mathbf{I}, \quad \bar{\mathbf{d}}_g = \text{sym } \bar{\mathbf{h}}_g = \mathbf{d}_g + q_g \mathbf{I}. \quad (105)$$

If  $\bar{\mathbf{c}}_g^{\text{T,alg}}$  is the Truesdell tangent obtained by pushing  $\bar{\mathbb{A}}_g^{\text{alg}}$  forward using  $\bar{\mathbf{F}}_g$  and  $J_c$ , the consistent directional derivative of Eq. (90) is

$$D\bar{G}^{e,\text{int}}[\Delta \mathbf{u}] = \int_{\Omega_e} \nabla_x \mathbf{v} : \left[ \bar{\mathbf{c}}^{\text{T,alg}} : \bar{\mathbf{d}} + \mathbf{h}\bar{\boldsymbol{\sigma}} - q\bar{\boldsymbol{\sigma}} \right] dv. \quad (106)$$

The three visible pieces are a projected material term, the ordinary geometric term, and an additional projection correction. The split is an implementation organization; Eq. (98) is the compact rate-neutral statement of the same tangent.

For each trial degree of freedom  $(B, k)$ , the projection scalar has the especially simple spatial form

$$q_g^{Bk} = \frac{1}{3} [(\nabla_x N_B|_c)_k - (\nabla_x N_B|_g)_k]. \quad (107)$$

Define a projected strain-displacement matrix column by

$$\bar{\mathbf{B}}_B^g(:, k) = \mathbf{B}_B^g(:, k) + q_g^{Bk} [1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0]^\top. \quad (108)$$

Here  $\mathbf{B}_B^g \in \mathbb{R}^{6 \times 3}$  is the usual node- $B$  block of the strain-displacement matrix at Gauss point  $g$ ,  $k \in \{1, 2, 3\}$  selects one of its displacement-component columns, and  $(:, k)$  denotes that column. Equivalently, the three columns define the projected block  $\bar{\mathbf{B}}_B^g \in \mathbb{R}^{6 \times 3}$ . This is finite-dimensional matrix notation, not a continuum tensor index.

Then the material part can be assembled as

$$\mathbf{K}_{\text{mat}}^{\bar{F}} = \sum_g \mathbf{B}_g^\top \bar{\mathbf{D}}_g^{\text{T,alg}} \bar{\mathbf{B}}_g \omega_g^x, \quad (109)$$

while the standard geometric block is

$$\mathbf{K}_{\text{geo}, AB}^{\bar{F}} = \sum_g \left[ (\nabla_x N_A)_g^\top \bar{\boldsymbol{\sigma}}_g (\nabla_x N_B)_g \right] \mathbf{I}_3 \omega_g^x. \quad (110)$$

Finally, with

$$\mathbf{q}_{B,g} = \frac{1}{3} (\nabla_x N_B|_c - \nabla_x N_B|_g), \quad (111)$$

the projection-correction block is

$$\mathbf{K}_{\text{proj}, AB}^{\bar{F}} = - \sum_g (\bar{\boldsymbol{\sigma}}_g \nabla_x N_A|_g) \otimes \mathbf{q}_{B,g} \omega_g^x. \quad (112)$$

Thus

$$\mathbf{K}_e^{\bar{F}} = \mathbf{K}_{\text{mat}}^{\bar{F}} + \mathbf{K}_{\text{geo}}^{\bar{F}} + \mathbf{K}_{\text{proj}}^{\bar{F}}. \quad (113)$$

The third term is caused by the nonlocal volumetric projection; it is not an additional material constitutive tangent returned by the constitutive update.

## Part IV

# Nonlinear solution method

## 16 Pseudo-time discretization and increment control

Pseudo-time  $t$  is a monotonically increasing parameter used to trace the prescribed loading path. The equilibrium equations are quasi-static, so  $t$  need not represent physical time for a rate-independent

material. A rate-dependent constitutive law may, however, use the physical increment  $\Delta t = t_{n+1} - t_n$  explicitly.

The nonlinear problem is advanced through accepted states

$$0 = t_0 < t_1 < \cdots < t_n < t_{n+1} \leq t_{\text{end}}. \quad (114)$$

The loading and prescribed kinematic data are evaluated at the target endpoint  $t_{n+1}$ . If a trial increment fails to converge, all trial quantities are discarded and the same advance from the committed state at  $t_n$  is attempted with a smaller  $\Delta t$ . If the step converges, the endpoint becomes the new committed state.

For path-dependent constitutive behavior, increments should not cross discontinuities or kinks in the prescribed loading path without explicitly landing on them. The concrete event schedule, minimum/-maximum step sizes, output synchronization, and cutback/growth policy are implementation-control choices and are specified separately in the implementation document.

## 17 Gauss-point committed and trial state

At each Gauss point the solver stores a model-owned committed constitutive state

$$\mathcal{H}_n^g. \quad (115)$$

The contents and length of  $\mathcal{H}$  are defined by the selected constitutive model. Examples include plastic deformation gradients, slip resistances, accumulated slips, evolving lattice orientation, hardening variables, or any other history required by the finite-increment update.

Geometry is kept separate from constitutive state. The committed deformation gradient  $\mathbf{F}_n^g$  is reconstructed from the reference coordinates and committed nodal displacement  $\mathbf{u}_n$  whenever the element is evaluated. Stress and tangent are material responses, not independent constitutive history unless a particular material model explicitly includes such quantities in its own state. The accepted Gauss-point Cauchy stress is nevertheless retained as a response cache for reporting and accepted-state equilibrium checks. This cache does not enter the constitutive integration map.

### 17.1 Observation is not a zero-duration constitutive increment

The supported cold start is the stress-free reference state:  $\mathbf{u}_0 = \mathbf{0}$ ,  $\mathbf{F}_0 = \mathbf{I}$ , and  $\boldsymbol{\sigma}_0^g = \mathbf{0}$ ; history is supplied by material initialization. A restart restores accepted nodal and constitutive data together with the accepted stress response. Reporting either state reconstructs kinematics and samples the accepted stress and history without invoking the evolution map. The internal-force residual can likewise be reconstructed from the accepted stress and geometry. No material or global tangent is required for this observation, and the saved stress does not replace trial stress in subsequent Newton assemblies.

For the first correction of the next increment,  $t_{n+1} > t_n$  even when the initial guess is  $\mathbf{u}_{n+1}^{(0)} = \mathbf{u}_n$  and hence  $\mathbf{F}_{n+1}^{g,(0)} = \mathbf{F}_n^g$ . Evaluate the material over this positive interval and differentiate its endpoint map with respect to the trial endpoint. Thus an algorithmic tangent is available before the first correction without a separate zero-duration evolution solve. This distinction also allows rate-dependent evolution at fixed deformation. The previous increment's tangent need not be saved: both Newton variants refresh a tangent at the beginning of each new attempt.

At global Newton iteration  $i$  for the fixed target pseudo-time  $t_{n+1}$ , the element constructs the trial kinematics and calls the native material update using the same  $\mathcal{H}_n^g$  on every global iteration. It receives a provisional state

$$\hat{\mathcal{H}}_{n+1}^{g,(i)} \quad (116)$$

that exists only in a trial buffer. No trial state is used as the starting state of the next global Newton iteration. Only after global convergence is

$$\mathcal{H}_{n+1}^g \leftarrow \widehat{\mathcal{H}}_{n+1}^{g,(i)} \quad (117)$$

committed atomically for all Gauss points.

## 17.2 Finite-increment constitutive integration is a separate local problem

Writing equilibrium at  $t_n$  and  $t_{n+1}$  does not define the material integration algorithm. A continuous rate description such as

$$\overset{\text{T}}{\boldsymbol{\sigma}} = \mathbb{C}^{\text{T}} : \mathbf{d} \quad (118)$$

does not by itself provide the discrete material map required by the finite element code. The local constitutive implementation must instead construct

$$(\mathbf{F}_n, \mathcal{H}_n) \longrightarrow (\mathbf{P}_{n+1}, \mathcal{H}_{n+1}) \quad (119)$$

using an appropriate finite-increment algorithm, for example a direct hyperelastic evaluation, implicit return mapping, exponential-map update, or crystal-plasticity solve. Its total derivative with respect to the trial endpoint is the algorithmic tangent  $\mathbb{A}^{\text{alg}}$ .

## 17.3 Geometry and local-failure classification

The solver distinguishes model errors from recoverable trial failures. A nonpositive or numerically degenerate reference mapping  $\det \mathbf{J}_0 \leq 0$  is a fatal mesh/model error. A committed configuration that cannot reconstruct a valid positive- $J$  deformation is likewise fatal because cutback cannot repair already accepted history. By contrast, if the current Newton trial  $\Omega_{n+1}^{(i)}$  produces  $\det \mathbf{J}_x \leq 0$ ,  $J = \det \mathbf{F} \leq 0$ , an invalid  $\bar{F}$  projection, or a recoverable local constitutive failure, the current increment attempt is abandoned and may be retried from the unchanged committed state with a cutback. Repeated failure is handled by the increment-control policy; if no admissible smaller increment remains, the analysis terminates.

# 18 Global nonlinear solution with exact linear displacement constraints

The element kernels developed above return unconstrained element residual and tangent contributions. General prescribed displacements, multipoint constraints, and periodic RVE kinematics are imposed only after global assembly. To avoid elimination of dependent displacement degrees of freedom, the selected global formulation uses Lagrange multipliers and solves the displacement and constraint equations simultaneously as a sparse saddle-point system.

## 18.1 General affine displacement constraints

Let the assembled displacement vector contain  $N$  displacement degrees of freedom,

$$\mathbf{u} \in \mathbb{R}^N, \quad (120)$$

and let  $m$  independent linear kinematic constraints be written as

$$\boxed{\mathbf{C}\mathbf{u} = \mathbf{d}}, \quad (121)$$

where

$$\mathbf{C} \in \mathbb{R}^{m \times N}, \quad \mathbf{d} \in \mathbb{R}^m. \quad (122)$$

The matrix  $\mathbf{C}$  is normally extremely sparse. A prescribed displacement  $u_j = \bar{u}_j$  requires a row with a single unit entry. A general multipoint constraint

$$\sum_{a=1}^{n_c} c_a u_{j_a} = \bar{d} \quad (123)$$

requires only  $n_c$  nonzero coefficients in that row.

For a periodic pair of reference points  $\mathbf{X}^+$  and  $\mathbf{X}^-$  under prescribed macroscopic deformation gradient  $\bar{\mathbf{F}}$ , the displacement-periodicity condition is

$$\boxed{\mathbf{u}^+ - \mathbf{u}^- = (\bar{\mathbf{F}} - \mathbf{I})(\mathbf{X}^+ - \mathbf{X}^-).} \quad (124)$$

Each Cartesian component gives one row of  $\mathbf{C}$  with coefficients +1 and -1 in the paired displacement degrees of freedom. The right-hand side  $\mathbf{d}$  changes with the prescribed macroscopic loading, while  $\mathbf{C}$  can remain fixed when the node pairing and reference geometry are fixed. Periodic pair constraints alone leave a rigid translation mode; one reference-node displacement or an equivalent zero-mean condition must also be supplied.

Equivalently, write the microscopic displacement as

$$\mathbf{u}(\mathbf{X}) = (\bar{\mathbf{F}} - \mathbf{I})\mathbf{X} + \tilde{\mathbf{u}}(\mathbf{X}), \quad (125)$$

where  $\tilde{\mathbf{u}}$  is periodic on opposite faces. The opposite-face contributions of  $\tilde{\mathbf{u}}$  cancel in the boundary integral, giving

$$\boxed{\frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{F} \, dV_0 = \bar{\mathbf{F}}.} \quad (126)$$

For a homogeneous body the fluctuation may vanish and the exact solution may be affine, so that case mainly checks pairing and affine reproduction. A heterogeneous core-matrix body generally develops a nonzero periodic fluctuation and distinct phase fields while still satisfying Eq. (126); these three observations together form a more discriminating periodic-boundary-condition test.

The constraints are assumed to have independent rows. Redundant constraints can make the augmented system singular even if every individual equation is physically reasonable. Removing obvious rigid modes is necessary but is not, for a general nonsymmetric tangent, a complete solvability test. The precise local condition is stated in Section 18.5 using the tangent restricted to the admissible displacement space.

## 18.2 Lagrange-multiplier equilibrium system

The unconstrained mechanical residual remains

$$\mathbf{R}(\mathbf{u}) = \mathbf{f}^{\text{int}}(\mathbf{u}) - \mathbf{f}^{\text{ext}}. \quad (127)$$

Introduce the multiplier vector

$$\lambda \in \mathbb{R}^m. \quad (128)$$

Using the sign convention adopted here, the constrained nonlinear equations are

$$\boxed{\mathcal{R}(\mathbf{u}, \lambda) = \begin{bmatrix} \mathbf{R}(\mathbf{u}) + \mathbf{C}^\top \lambda \\ \mathbf{C}\mathbf{u} - \mathbf{d} \end{bmatrix} = \mathbf{0}.} \quad (129)$$

The nodal force exerted by the constraints on the displacement system is, with this sign convention,

$$\mathbf{f}^{\text{con}} = -\mathbf{C}^\top \lambda. \quad (130)$$

Thus the invariant physical constraint-force vector is  $\mathbf{f}^{\text{con}} = -\mathbf{C}^\top \lambda$ . Individual multiplier components depend on the chosen scaling and basis of the constraint rows: if one equation is multiplied by a nonzero scalar, its multiplier rescales inversely while  $\mathbf{C}^\top \lambda$  is unchanged. For periodic constraints, different but equivalent spanning trees likewise produce different multiplier coordinates. Reactions and homogenization diagnostics should therefore be based on assembled nodal constraint forces (or quantities derived from them), not on raw multiplier values alone. For a single retained periodic pair row, the structure of  $\mathbf{C}^\top \lambda$  produces equal and opposite nodal pair forces.

For a periodic cell with no body force or separately applied nodal load, equilibrium gives  $\mathbf{f}^{\text{int}} = \mathbf{f}^{\text{con}}$ . Isoparametric completeness then provides a convenient macroscopic nominal-stress diagnostic directly from the nodal constraint reactions:

$$\boxed{\bar{\mathbf{P}} = \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P} \, dV_0 = \frac{1}{|\Omega_0|} \sum_a \mathbf{f}_a^{\text{con}} \otimes \mathbf{X}_a.} \quad (131)$$

Indeed,  $\sum_a \mathbf{f}_a^{\text{int}} \otimes \mathbf{X}_a$  reduces to the volume integral of  $\mathbf{P}$  because  $\sum_a \mathbf{X}_a \otimes \text{Grad } N_a = \mathbf{I}$ . Equation (131) uses the complete assembled nodal reaction vector, including the anchor reaction; it does not assign physical meaning to individual multiplier coordinates. If body forces or independent nodal loads are present, their force moments must be included and this reduced expression cannot be used unchanged.

Because the constraints are linear and are taken to be independent of the unknown displacement within an increment, the Newton Jacobian is

$$\boxed{\mathbf{K}_{\text{KKT}}^{(i)} = \begin{bmatrix} \mathbf{K}^{(i)} & \mathbf{C}^\top \\ \mathbf{C} & \mathbf{0} \end{bmatrix}}, \quad (132)$$

where

$$\mathbf{K}^{(i)} = \left. \frac{\partial \mathbf{f}^{\text{int}}}{\partial \mathbf{u}} \right|_{\mathbf{u}^{(i)}} \quad (133)$$

contains the assembled material, geometric, and  $\bar{F}$  projection terms appropriate to the selected elements.

Distinguish this current Jacobian from the tangent selected for the solve:

$$\hat{\mathbf{K}}^{(i)} = \begin{cases} \mathbf{K}^{(i)}, & \text{exact (full) Newton,} \\ \mathbf{K}^{(0)}, & \text{modified Newton.} \end{cases} \quad (134)$$

The modified-Newton tangent and its KKT factorization are frozen for one increment *attempt*, not across cutbacks. Its first tangent is evaluated at  $\mathbf{u}^{(0)} = \mathbf{u}_n$  for the positive duration  $t_{n+1} - t_n$ ; it is not a saved tangent evaluated at zero duration at  $t_n$ .

At Newton iteration  $i$  the simultaneous correction is obtained from

$$\boxed{\begin{bmatrix} \hat{\mathbf{K}}^{(i)} & \mathbf{C}^\top \\ \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{u}^{(i)} \\ \Delta \lambda^{(i)} \end{bmatrix} = - \begin{bmatrix} \mathbf{f}^{\text{int},(i)} - \mathbf{f}^{\text{ext}} + \mathbf{C}^\top \lambda^{(i)} \\ \mathbf{C} \mathbf{u}^{(i)} - \mathbf{d} \end{bmatrix}}. \quad (135)$$

The undamped updates are

$$\mathbf{u}^{(i+1)} = \mathbf{u}^{(i)} + \Delta \mathbf{u}^{(i)}, \quad (136)$$

$$\lambda^{(i+1)} = \lambda^{(i)} + \Delta \lambda^{(i)}. \quad (137)$$

No constrained displacement is eliminated, and  $\mathbf{K}$  is not separately factorized. The complete sparse matrix of Eq. (135) is supplied to the linear solver.

### 18.3 Convergence and residual-based backtracking

At the current iterate, the absolute-plus-relative tolerances are

$$\begin{aligned}\epsilon_{f,i} &= \epsilon_f^{\text{abs}} + \epsilon_f^{\text{rel}} \max(\|\mathbf{f}^{\text{int},(i)}\|_\infty, \|\mathbf{f}^{\text{ext}}\|_\infty, \|\mathbf{C}^\top \lambda^{(i)}\|_\infty), \\ \epsilon_{c,i} &= \epsilon_c^{\text{abs}} + \epsilon_c^{\text{rel}} \max(\|\mathbf{C}\mathbf{u}^{(i)}\|_\infty, \|\mathbf{d}\|_\infty).\end{aligned}\tag{138}$$

Both residual tests,  $\|\mathbf{r}_u^{(i)}\|_\infty \leq \epsilon_{f,i}$  and  $\|\mathbf{r}_c^{(i)}\|_\infty \leq \epsilon_{c,i}$ , are mandatory. An optional supplementary test uses the *last accepted displacement correction*

$$\begin{aligned}\mathbf{s}^{(0)} &= \mathbf{0}, \quad \mathbf{s}^{(i)} = \mathbf{u}^{(i)} - \mathbf{u}^{(i-1)} \quad (i > 0), \\ \|\mathbf{s}^{(i)}\|_\infty &\leq \epsilon_u^{\text{abs}} + \epsilon_u^{\text{rel}} \max(\|\mathbf{u}^{(i)}\|_\infty, \|\mathbf{u}_n\|_\infty).\end{aligned}\tag{139}$$

Thus the two convergence policies are residual-only, or residuals *and* a small accepted correction; they are independent of the choice of Newton method. The supplied examples enable the supplementary test by setting `check_displacement_increment` to `true`. If the option is absent or false, only the residual tests apply. A small correction alone never establishes equilibrium.

When globalization is required, the same KKT solution is retained as the Newton search direction but the update is damped,

$$\mathbf{u}^{(i+1)} = \mathbf{u}^{(i)} + \alpha_i \Delta \mathbf{u}^{(i)},\tag{140}$$

$$\lambda^{(i+1)} = \lambda^{(i)} + \alpha_i \Delta \lambda^{(i)}, \quad 0 < \alpha_i \leq 1.\tag{141}$$

This does not turn exact Newton into modified Newton: exact Newton still forms the tangent at every accepted trial iterate. It changes only the length taken along the current Newton direction.

For the residual-based backtracking used in the necking examples, define the dimensionless merit at the base iterate as

$$M_i = \max \left( \frac{\|\mathbf{r}_u^{(i)}\|_\infty}{\epsilon_{f,i}}, \frac{\|\mathbf{r}_c^{(i)}\|_\infty}{\epsilon_{c,i}} \right),\tag{142}$$

The same base-iterate tolerances  $\epsilon_{f,i}$  and  $\epsilon_{c,i}$  are held fixed for every candidate in this line search; they are reevaluated at the next Newton iterate. For the candidate pair  $(\mathbf{u}_\alpha, \lambda_\alpha) = (\mathbf{u}^{(i)}, \lambda^{(i)}) + \alpha(\Delta \mathbf{u}^{(i)}, \Delta \lambda^{(i)})$ , define

$$M(\alpha) = \max \left( \frac{\|\mathbf{r}_u(\mathbf{u}_\alpha, \lambda_\alpha)\|_\infty}{\epsilon_{f,i}}, \frac{\|\mathbf{r}_c(\mathbf{u}_\alpha)\|_\infty}{\epsilon_{c,i}} \right).\tag{143}$$

If a tolerance is zero, its ratio is interpreted as zero for a zero residual and infinity otherwise. Starting with  $\alpha_i = 1$ , reduce it geometrically until the candidate is kinematically admissible and

$$M(\alpha_i) \leq (1 - c\alpha_i)M_i, \quad 0 \leq c < 1.\tag{144}$$

Every candidate constitutive update is evaluated from the same committed pair  $(\mathbf{F}_n, \mathcal{H}_n)$ , exactly as for an ordinary Newton trial. Hence a rejected candidate cannot contaminate history. A nonpositive trial Jacobian or another recoverable local failure rejects that search length. Backtracking tests  $\alpha = 1, \beta, \beta^2, \dots$  with  $0 < \beta < 1$ , bounded by a minimum length  $\alpha_{\min}$  and at most  $B_{\max}$  reductions. Failure to find an admissible decreasing length within these limits is a recoverable increment failure and invokes cutback. Fatal model/setup errors are not repaired by backtracking.

The merit test requires stresses and trial state, not their endpoint derivative. A residual-only evaluation may therefore omit constitutive differentiation and stiffness construction without changing the integration map. The current exact-Newton implementation nevertheless constructs each candidate tangent, including rejected candidates, so that the accepted assembly can be reused at the next iteration. Modified-Newton candidates omit it and retain the frozen tangent. Both methods cache the accepted candidate's stress, trial state, and internal-force assembly; exact Newton also caches its tangent. The next iteration consumes this assembly instead of reintegrating the same endpoint. A line-search acceptance is not a material-state commitment. The supplementary correction test uses  $\mathbf{s}^{(i+1)} = \alpha_i \Delta \mathbf{u}^{(i)}$ , not the undamped direction.



## 18.4 Local error analysis of the search direction

Let  $\mathbf{x} = (\mathbf{u}^\top, \lambda^\top)^\top$ ,  $\mathbf{r} = \mathcal{R}(\mathbf{x})$ , and  $\mathbf{J}^{(i)} = D\mathcal{R}(\mathbf{x}^{(i)})$  be the *actual* KKT Jacobian. For a smooth residual with a locally Lipschitz Jacobian, and fixed committed history and target time,

$$\mathbf{r}(\mathbf{x}^{(i)} + \alpha \Delta \mathbf{x}^{(i)}) = \mathbf{r}^{(i)} + \alpha \mathbf{J}^{(i)} \Delta \mathbf{x}^{(i)} + \mathcal{O}(\alpha^2 \|\Delta \mathbf{x}^{(i)}\|^2). \quad (145)$$

For exact Newton with an accurately solved correction,  $\mathbf{J}^{(i)} \Delta \mathbf{x}^{(i)} = -\mathbf{r}^{(i)}$ , hence

$$\boxed{\mathbf{r}(\alpha) = (1 - \alpha) \mathbf{r}^{(i)} + \mathcal{O}(\alpha^2 \|\Delta \mathbf{x}^{(i)}\|^2).} \quad (146)$$

At fixed direction, halving  $\alpha$  approximately quarters the leading quadratic *remainder*, not the total residual: the retained term approaches  $\mathbf{r}^{(i)}$  as  $\alpha \rightarrow 0$ . Near a regular solution, a full step cancels that first-order term and is typically acceptable. This is a local smoothness argument, not a guarantee of success within finite backtracking limits or across nonsmooth constitutive transitions.

For modified Newton, the solve instead uses  $\widehat{\mathbf{J}}^{(i)}$  with mechanical block  $\mathbf{K}^{(0)}$ . Substitution gives

$$\boxed{\begin{aligned} \mathbf{r}(\alpha) &= (1 - \alpha) \mathbf{r}^{(i)} + \alpha (\mathbf{J}^{(i)} - \widehat{\mathbf{J}}^{(i)}) \Delta \mathbf{x}^{(i)} \\ &\quad + \mathcal{O}(\alpha^2 \|\Delta \mathbf{x}^{(i)}\|^2). \end{aligned}} \quad (147)$$

The Jacobian-mismatch term is *first order* in  $\alpha$ ; reducing the search length does not eliminate it relative to the intended first-order residual reduction. A frozen-tangent direction need not be a descent direction for the chosen merit, so arbitrarily small steps need not satisfy the acceptance test. Backtracking can control such directions but cannot guarantee their success. Approximate material tangents or inexact linear solves likewise weaken the exact-Newton cancellation.

Both methods retain the exact linear constraint block, so their accurately solved corrections satisfy

$$\mathbf{C} \Delta \mathbf{u}^{(i)} = -\mathbf{r}_c^{(i)}, \quad \mathbf{r}_c(\alpha) = (1 - \alpha) \mathbf{r}_c^{(i)}. \quad (148)$$

A full step therefore removes the constraint residual up to roundoff, even in modified Newton; damping contracts it rather than eliminating it immediately.

For the linear special case  $\mathbf{R} = \mathbf{K} \mathbf{u} - \mathbf{f}$ , Eq. (135) reduces directly to

$$\boxed{\begin{bmatrix} \mathbf{K} & \mathbf{C}^\top \\ \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{d} \end{bmatrix}.} \quad (149)$$

This is the standard Lagrange-multiplier saddle-point or KKT form for affine displacement constraints.

An important property of linear constraints is that the second block row of Eq. (135) gives

$$\mathbf{C} \Delta \mathbf{u}^{(i)} = -(\mathbf{C} \mathbf{u}^{(i)} - \mathbf{d}). \quad (150)$$

Therefore, after one accurately solved *undamped* Newton correction the trial displacement satisfies the linear constraints to linear-solver tolerance. With Eq. (141), the constraint residual is instead multiplied by  $1 - \alpha_i$  and is removed over subsequent accepted corrections. Once the constraints are satisfied, subsequent corrections within the same increment satisfy  $\mathbf{C} \Delta \mathbf{u} \simeq \mathbf{0}$  because  $\mathbf{d}$  is fixed at the target pseudo-time.

## 18.5 Indefiniteness, solvability, and physical relevance

The KKT matrix in Eq. (132) is normally *indefinite*; this is not the same as being indeterminate or singular. The zero multiplier block introduces positive and negative algebraic directions even when the constrained mechanical problem has a unique stable solution.

Assume  $\mathbf{C}$  has full row rank and let the columns of  $\mathbf{Z}$  form a basis of the admissible homogeneous displacement space,

$$\mathbf{C}\mathbf{Z} = \mathbf{0}. \quad (151)$$

Then the local constrained Newton problem has a unique displacement correction if and only if the reduced tangent

$$\boxed{\mathbf{K}_{\text{red}} = \mathbf{Z}^\top \mathbf{K} \mathbf{Z}} \quad (152)$$

is nonsingular. This statement applies also when  $\mathbf{K}$  is nonsymmetric. The easier diagnostic  $\ker(\mathbf{K}) \cap \ker(\mathbf{C}) = \{\mathbf{0}\}$  is necessary in common cases but is not used as the general sufficient condition here.

For a symmetric conservative problem,  $\mathbf{K}_{\text{red}} \succ 0$  is the familiar local stability condition. The KKT matrix can remain indefinite while this reduced tangent is positive definite. Conversely, loss of rank or definiteness of the reduced mechanical tangent can signal a genuine instability rather than a boundary-condition error. These are *local* statements: a nonsingular tangent does not exclude the existence of multiple remote nonlinear equilibrium branches.

A singular factorization must therefore be diagnosed rather than interpreted automatically as nonlinear nonconvergence. Common setup causes are an unremoved rigid mode, linearly dependent rows of  $\mathbf{C}$ , or inconsistent constraints. Mechanical causes include an invalid/degenerate element and an actual loss of rank associated with material or structural instability. For the small verification problems, a debug mode may check the numerical rank of  $\mathbf{C}$  and, for symmetric cases, the smallest eigenvalues of a reduced tangent. These checks are diagnostic only and are not part of the production sparse solve.

## 18.6 Sparse storage and linear solution strategy

The Lagrange-multiplier formulation preserves the sparsity inherited from finite element assembly, but it changes the matrix class. Even when  $\mathbf{K}$  is symmetric positive definite on unconstrained displacement space, the zero multiplier block makes the KKT matrix indefinite. If  $\mathbf{K}$  is nonsymmetric, as is permitted here, the complete KKT matrix is generally nonsymmetric and indefinite.

The baseline solution strategy is therefore a sparse factorization or sparse linear solver that accepts a general indefinite matrix. No separate factorization of the unconstrained  $\mathbf{K}$  is required. For symmetric problems, symmetric-indefinite methods are possible; for general nonsymmetric problems, a nonsymmetric sparse direct method or a suitably preconditioned Krylov method is required. A separate implementation specification selects concrete libraries and storage formats.

For exact Newton,  $\mathbf{K}^{(i)}$  changes and the KKT operator is normally rebuilt/refactorized at each iteration. For modified Newton, the selected tangent may be held fixed while stresses, residuals, and trial constitutive states are still recomputed at every displacement iterate.

The multiplier equations and displacement equations have different physical units, and arbitrary multipoint coefficients can differ greatly in magnitude. Rescaling one constraint row and the corresponding entry of  $\mathbf{d}$  by the same nonzero scalar leaves the admissible displacement set unchanged but rescales its multiplier. Sensible row scaling is therefore a conditioning device; physical reactions should be interpreted through  $-\mathbf{C}^\top \lambda$ , which is invariant under such row rescaling.

## 18.7 Relation between global Newton and the local material tangent

The local material-point update remains exactly as developed in the preceding sections. At global iteration  $i$ , every element is evaluated using the current trial nodal displacement  $\mathbf{u}_{n+1}^{(i)}$  but the committed Gauss-point state at  $t_n$ . The element returns its trial state, internal force, and consistent element tangent. Global assembly produces

$$\mathbf{f}^{\text{int},(i)}, \quad \mathbf{K}^{(i)}. \quad (153)$$

The constraint blocks do not alter the local constitutive update. They enter only after assembly through Eq. (135).

A consistent local algorithmic tangent makes the assembled mechanical block  $\mathbf{K}$  consistent with the actual discrete constitutive update. This supports quadratic convergence when the global residual is smooth, the active set is stable, the full tangent is rebuilt, and the linear KKT system is solved sufficiently accurately. The Lagrange-multiplier constraints do not require a different material tangent.

A modified Newton method holds  $\mathbf{K}$  fixed over the remaining iterations of an attempt. This does not alter material state logic: at every distinct trial displacement the material update and internal force are still evaluated from the committed  $t_n$  state. Consuming an accepted candidate assembly avoids repeating the same evaluation. Only tangent assembly/factorization is skipped. The first correction of every new attempt, including a cutback retry, uses a freshly formed tangent/factorization.

## Part V

# Element and global algorithms

## 19 Element algorithm: standard full-integration formulation

The algorithm below covers both standard full-integration element formulations: Hex8, Hex20. The only element-type-specific inputs are shape functions, derivatives, node count, and quadrature. It starts from reference/committed/trial nodal data and committed Gauss-point states and returns the element internal force, tangent, and trial Gauss-point states. It contains no global assembly or boundary-condition enforcement. All spatial gradients, current quadrature volumes, Cauchy stresses, and spatial tangent matrices in an iteration are associated with the trial geometry  $\Omega_{n+1}^{(i)}$ ; committed geometry is used only to reconstruct the beginning-of-increment kinematics supplied to the material update.

### Inputs

- element formulation  $e_{\text{type}} \in \{\text{hex8}, \text{hex20}\}$  and its shape-function/quadrature tables;
- reference nodal coordinates  $\{\mathbf{X}_A\}_{A=1}^{n_e}$ ;
- committed nodal displacement  $\{\mathbf{u}_{A,n}\}$ ;
- current trial nodal displacement  $\{\mathbf{u}_{A,n+1}^{(i)}\}$ ;
- committed Gauss-point states  $\{\mathcal{H}_n^g\}$ ;
- constitutive model data and pseudo-time endpoints  $(t_n, t_{n+1})$ .

### Outputs

- $\mathbf{f}^{e,\text{int}}$ ;
- $\mathbf{K}_e$ ;
- trial histories  $\{\widehat{\mathcal{H}}_{n+1}^{g,(i)}\}$ ;
- optional element/material status requesting a global cutback.

## Procedure

1. Set

$$\mathbf{f}^{e,\text{int}} = \mathbf{0}, \quad \mathbf{K}_{\text{mat}} = \mathbf{0}, \quad \mathbf{K}_{\text{geo}} = \mathbf{0}.$$

2. Construct the committed and current trial nodal positions

$$\mathbf{x}_{A,n} = \mathbf{X}_A + \mathbf{u}_{A,n}, \quad \mathbf{x}_{A,n+1}^{(i)} = \mathbf{X}_A + \mathbf{u}_{A,n+1}^{(i)}.$$

3. For every Gauss point of the selected full-integration rule:

- (a) Evaluate  $N_A(\boldsymbol{\xi}_g)$  and  $\nabla_{\boldsymbol{\xi}} N_A(\boldsymbol{\xi}_g)$ .

- (b) Compute the reference, committed, and current-trial element Jacobians

$$\mathbf{J}_{0,g}, \quad \mathbf{J}_{x,n,g}, \quad \mathbf{J}_{x,n+1,g}^{(i)}.$$

Reject or flag the element if an admissible positive Jacobian condition is violated.

- (c) Compute

$$\nabla_X N_A = \mathbf{J}_0^{-\top} \nabla_{\boldsymbol{\xi}} N_A, \quad \nabla_x N_A = (\mathbf{J}_{x,n+1}^{(i)})^{-\top} \nabla_{\boldsymbol{\xi}} N_A.$$

- (d) Compute both endpoint deformation gradients from geometry,

$$\mathbf{F}_n^g = \mathbf{J}_{x,n,g} \mathbf{J}_{0,g}^{-1}, \quad \mathbf{F}_{n+1}^{g,(i)} = \mathbf{J}_{x,n+1,g}^{(i)} \mathbf{J}_{0,g}^{-1}, \quad J_{n+1}^{g,(i)} = \det \mathbf{F}_{n+1}^{g,(i)}.$$

- (e) Evaluate the discrete material map

$$\mathcal{M} \left( \mathbf{F}_n^g, \mathbf{F}_{n+1}^{g,(i)}, \mathcal{H}_n^g, t_n, t_{n+1} \right) \longrightarrow \left( \mathbf{P}_{n+1}^{g,(i)}, \mathbb{A}_{n+1,g}^{\text{alg},(i)}, \widehat{\mathcal{H}}_{n+1}^{g,(i)} \right).$$

If the local constitutive update has no admissible solution at the trial endpoint, return a recoverable failure without modifying committed history.

- (f) Convert the returned first Piola stress to Cauchy stress:

$$\boldsymbol{\sigma}_{n+1}^{g,(i)} = \frac{1}{J_{n+1}^{g,(i)}} \mathbf{P}_{n+1}^{g,(i)} (\mathbf{F}_{n+1}^{g,(i)})^\top.$$

- (g) Push  $\mathbb{A}^{\text{alg}}$  forward using Eqs. (34) and (37) to obtain  $\mathbb{C}_g^{\text{T,alg}}$ , then convert it consistently to the  $6 \times 6$  matrix  $\mathbf{D}_g^{\text{T,alg}}$  using the engineering-shear convention.

- (h) Construct the current  $\mathbf{B}_g$  matrix from  $\nabla_x N_A$ .

- (i) Define the current quadrature volume

$$\omega_g^x = \det \mathbf{J}_{x,n+1,g}^{(i)} W_g.$$

- (j) Accumulate internal force

$$\mathbf{f}^{e,\text{int}} += \mathbf{B}_g^\top \mathbf{s}_g \omega_g^x.$$

- (k) Accumulate material tangent

$$\mathbf{K}_{\text{mat}} += \mathbf{B}_g^\top \mathbf{D}_g^{\text{T,alg}} \mathbf{B}_g \omega_g^x.$$

- (l) For each node pair  $A, B$ , accumulate the  $3 \times 3$  geometric block

$$\mathbf{K}_{\text{geo},AB} += \left[ (\nabla_x N_A)^\top \boldsymbol{\sigma}_g \nabla_x N_B \right] \mathbf{I}_3 \omega_g^x.$$

- (m) Store  $\widehat{\mathcal{H}}_{n+1}^{g,(i)}$  as a trial result only.

4. Form

$$\boxed{\mathbf{K}_e = \mathbf{K}_{\text{mat}} + \mathbf{K}_{\text{geo}}}.$$

### Final element kernel – standard full-integration element

**Input:**  $\{\mathbf{X}_A\}$ , committed  $\{\mathbf{u}_{A,n}\}$ , trial  $\{\mathbf{u}_{A,n+1}^{(i)}\}$ , committed Gauss-point state  $\{\mathcal{H}_n^g\}$ , resolved material/property data,  $(t_n, t_{n+1})$ .

1. Form both  $\mathbf{x}_{A,n} = \mathbf{X}_A + \mathbf{u}_{A,n}$  and  $\mathbf{x}_{A,n+1}^{(i)} = \mathbf{X}_A + \mathbf{u}_{A,n+1}^{(i)}$ ; zero  $\mathbf{f}^{e,\text{int}}$ ,  $\mathbf{K}_{\text{mat}}$ ,  $\mathbf{K}_{\text{geo}}$ .
2. For each Gauss point ( $n_g = 8$  for Hex8 and  $n_g = 27$  for Hex20), evaluate  $N_A$ ,  $\nabla_\xi N_A$ ,  $\mathbf{J}_0$ , committed/trial  $\mathbf{J}_x$ ,  $\nabla_X N_A$ , trial  $\nabla_x N_A$ ,  $\mathbf{F}_n = \mathbf{J}_{x,n} \mathbf{J}_0^{-1}$ ,  $\mathbf{F}_{n+1}^{(i)} = \mathbf{J}_{x,n+1}^{(i)} \mathbf{J}_0^{-1}$ , and  $J = \det \mathbf{F}_{n+1}^{(i)}$ .
3. From the same committed state at  $n$ , evaluate the discrete material map  $\mathcal{M}$  from  $(\mathbf{F}_n^g, \mathbf{F}_{n+1}^{g,(i)}, \mathcal{H}_n^g, t_n, t_{n+1})$  and obtain  $(\mathbf{P}_{n+1}^{g,(i)}, \mathbb{A}_g^{\text{alg}}, \widehat{\mathcal{H}}_{n+1}^{g,(i)})$ .
4. If the local update fails, return a cutback request. Otherwise set  $\boldsymbol{\sigma} = J^{-1} \mathbf{P} \mathbf{F}^\top$ .
5. Push  $\mathbb{A}^{\text{alg}}$  forward with Eqs. (34)–(37) to obtain  $\mathbf{D}^{\text{T,alg}}$ ; form  $\mathbf{B}$  and  $\omega_g^x = \det(\mathbf{J}_x) W_g$ .
6. Accumulate  $\mathbf{f}^{e,\text{int}} += \mathbf{B}^\top \mathbf{s} \omega_g^x$ ,  $\mathbf{K}_{\text{mat}} += \mathbf{B}^\top \mathbf{D}^{\text{T,alg}} \mathbf{B} \omega_g^x$ , and  $\mathbf{K}_{\text{geo}}$  from Eq. (79).
7. Store the Gauss-point result as *trial* state only.

**Return:**

$$\boxed{\mathbf{K}_e^{\text{T}} = \mathbf{K}_{\text{mat}} + \mathbf{K}_{\text{geo}}}, \quad \mathbf{f}^{e,\text{int}}, \quad \{\widehat{\mathcal{H}}_{n+1}^{g,(i)}\}.$$

## 20 Element algorithm: centroidal $\bar{F}$ Hex8

This algorithm uses the same discrete constitutive map as the standard elements. The only additional element-level operations are the centroidal volumetric projection and the consistent chain rule derived in Section 15.4.

### Inputs and outputs

The inputs and outputs are identical to the standard Truesdell element algorithm, except that the element type is fixed to Hex8 with centroidal  $\bar{F}$ . In particular, the constitutive map returns the same mathematical response quantities

$$\{\mathbf{P}, \mathbb{A}^{\text{alg}}, \widehat{\mathcal{H}}_{n+1}\} \quad (154)$$

with the same meanings and tensor dimensions; only the deformation gradients supplied by the element wrapper differ.

### Final element kernel – centroidal $\bar{F}$ Hex8

**Input:** reference Hex8 coordinates  $\{\mathbf{X}_A\}_{A=1}^8$ , committed displacement  $\{\mathbf{u}_{A,n}\}$ , trial displacement  $\{\mathbf{u}_{A,n+1}^{(i)}\}$ , committed Gauss-point state  $\{\mathcal{H}_n^g\}_{g=1}^8$ , resolved material/property data,  $(t_n, t_{n+1})$ .

1. Form committed and trial nodes  $\mathbf{x}_{A,n} = \mathbf{X}_A + \mathbf{u}_{A,n}$  and  $\mathbf{x}_{A,n+1}^{(i)} = \mathbf{X}_A + \mathbf{u}_{A,n+1}^{(i)}$ . At  $\boldsymbol{\xi}_c = (0, 0, 0)^\top$ , evaluate reference, committed, and trial maps and compute  $\mathbf{F}_{c,n}$ ,  $J_{c,n}$ , trial  $\mathbf{F}_c$ ,  $J_c$ , and trial  $\nabla_x N_A|_c$ . No constitutive routine is called at the centroid.
2. Zero  $\mathbf{f}^{e,\text{int}}$ ,  $\mathbf{K}_{\text{mat}}^{\bar{F}}$ ,  $\mathbf{K}_{\text{geo}}^{\bar{F}}$ , and  $\mathbf{K}_{\text{proj}}^{\bar{F}}$ .
3. For each of the eight  $2 \times 2 \times 2$  Gauss points:
  - a. Compute  $\mathbf{J}_0$ ,  $\mathbf{J}_x$ ,  $\nabla_X N_A$ ,  $\nabla_x N_A$ , raw  $\mathbf{F}_g$ ,  $J_g$ , and the corresponding committed raw  $\mathbf{F}_{n,g}$ .
  - b. Form

$$\alpha_g = (J_c/J_g)^{1/3}, \quad \bar{\mathbf{F}}_g = \alpha_g \mathbf{F}_g,$$

and analogously  $\bar{\mathbf{F}}_{n,g} = \alpha_{n,g} \mathbf{F}_{n,g}$  from the committed configuration.

- c. Call the same native constitutive update used by a standard element,

$$\mathcal{M}(\bar{\mathbf{F}}_{n,g}, \bar{\mathbf{F}}_g, \mathcal{H}_n^g, t_n, t_{n+1}) \longrightarrow (\bar{\mathbf{P}}_g, \bar{\mathbb{A}}_g^{\text{alg}}, \hat{\mathcal{H}}_{n+1}^{g,(i)}).$$

If the local update fails, return a cutback request without committing state.

- d. Convert only with the deformation gradient seen by the material:

$$\bar{\boldsymbol{\sigma}}_g = J_c^{-1} \bar{\mathbf{P}}_g \bar{\mathbf{F}}_g^\top.$$

Do *not* construct the stress using raw  $\mathbf{F}_g$ .

- e. Push  $\bar{\mathbb{A}}_g^{\text{alg}}$  forward with  $\bar{\mathbf{F}}_g$  and  $J_c$  to obtain  $\bar{\mathbf{D}}_g^{\text{T,alg}}$ .

- f. Form  $\mathbf{B}_g$  from the raw current gradients and, for every node  $B$ , set

$$\mathbf{q}_{B,g} = \frac{1}{3} (\nabla_x N_B|_c - \nabla_x N_B|_g), \quad \bar{\mathbf{B}}_B^g(:, k) = \mathbf{B}_B^g(:, k) + q_{B,g,k} [1, 1, 1, 0, 0, 0]^\top.$$

- g. With  $\omega_g^x = \det \mathbf{J}_{x,g} W_g$ , accumulate

$$\mathbf{f}^{e,\text{int}} += \mathbf{B}_g^\top \bar{\mathbf{s}}_g \omega_g^x,$$

$$\mathbf{K}_{\text{mat}}^{\bar{F}} += \mathbf{B}_g^\top \bar{\mathbf{D}}_g^{\text{T,alg}} \bar{\mathbf{B}}_g \omega_g^x,$$

the ordinary geometric blocks using  $\bar{\boldsymbol{\sigma}}_g$ , and

$$\mathbf{K}_{\text{proj,AB}}^{\bar{F}} += -(\bar{\boldsymbol{\sigma}}_g \nabla_x N_A) \otimes \mathbf{q}_{B,g} \omega_g^x.$$

- h. Store the returned Gauss-point state as trial state only.

4. Return

$$\boxed{\mathbf{K}_e^{\bar{F}} = \mathbf{K}_{\text{mat}}^{\bar{F}} + \mathbf{K}_{\text{geo}}^{\bar{F}} + \mathbf{K}_{\text{proj}}^{\bar{F}}}$$

together with  $\mathbf{f}^{e,\text{int}}$ , the eight trial Gauss-point states.

**Equivalent direct- $DP/DF$  implementation.** For debugging, or as an alternative production kernel, steps 3(e)–3(g) may be replaced exactly by Eqs. (94) and (103). The two implementations must agree to numerical precision. This gives a strong element-level test of the spatial Truesdell conversion, the geometric term, and the  $\bar{F}$  projection correction simultaneously.

#### Final element kernel – standard Hex8 control

The standard Hex8 uses the ordinary Truesdell algorithm of Section 19 with  $n_e = 8$  and  $n_g = 8$ . At every Gauss point set  $\mathbf{F}^{\text{mat}} = \mathbf{F}$ , call the same constitutive update, assemble

$$\mathbf{f}^{e,\text{int}} = \sum_g \mathbf{B}_g^\top \mathbf{s}_g \omega_g^x, \quad \mathbf{K}_e = \sum_g \mathbf{B}_g^\top \mathbf{D}_g^{\text{T,alg}} \mathbf{B}_g \omega_g^x + \mathbf{K}_{\text{geo}},$$

and store only trial material states until global convergence. There is no centroid evaluation,  $\alpha_g = 1$ ,  $\bar{\mathbf{B}} = \mathbf{B}$ , and  $\mathbf{K}_{\text{proj}} = \mathbf{0}$ .

## 21 Global nonlinear solution algorithm

Algorithm 1 covers one increment *attempt* with backtracking enabled. Write  $\mathbf{x} = (\mathbf{u}, \lambda)$  and suppress the trial endpoint subscript  $n + 1$ . Every constitutive evaluation uses the same committed  $(\mathbf{u}_n, \mathcal{H}_n)$  and the current trial displacement, never a previous Newton trial history. Convergence requires both residual tests and, when enabled, the accepted-correction test of Section 18.3.

---

**Algorithm 1** Full or modified Newton with bounded backtracking: one increment attempt

---

**Require:** method, committed  $(\mathbf{x}_n, \mathcal{H}_n)$ , target  $t_{n+1} > t_n$ , iteration/search limits

```

1:  $\mathbf{x}^{(0)} \leftarrow \mathbf{x}_n, \mathbf{s}^{(0)} \leftarrow \mathbf{0}$ 
2: for  $i = 0, \dots, N_{\max}$  do
3:   Obtain assembly at  $\mathbf{u}^{(i)}$  (evaluate or reuse accepted candidate)
4:   Form  $\mathbf{r}_u^{(i)}, \mathbf{r}_c^{(i)}$  and current tolerances
5:   if converged then
6:     Commit  $\mathbf{x}^{(i)}$  and trial history; return success
7:   end if
8:   if  $i = N_{\max}$  then
9:     return recoverable failure
10:  end if
11:  if full Newton or  $i = 0$  then
12:     $\hat{\mathbf{K}} \leftarrow \mathbf{K}^{(i)}$ ; factor the KKT matrix
13:  end if
14:  Solve Eq. (135) for  $\Delta \mathbf{x}^{(i)}$ 
15:  Evaluate  $M_i$  (Eq. (142));  $\alpha \leftarrow 1$ ; accepted  $\leftarrow$  false
16:  for  $b = 0, \dots, B_{\max}$  do
17:    if  $\alpha < \alpha_{\min}$  then
18:      break
19:    end if
20:     $\mathbf{x}_\alpha \leftarrow \mathbf{x}^{(i)} + \alpha \Delta \mathbf{x}^{(i)}$ 
21:    Evaluate candidate assembly and  $M(\alpha)$ 
22:    if admissible and  $M(\alpha) \leq (1 - c\alpha)M_i$  then
23:      Cache candidate assembly; accepted  $\leftarrow$  true; break
24:    end if
25:     $\alpha \leftarrow \beta\alpha$ 
26:  end for
27:  if not accepted then
28:    return recoverable failure
29:  end if
30:   $\mathbf{x}^{(i+1)} \leftarrow \mathbf{x}_\alpha, \mathbf{s}^{(i+1)} \leftarrow \alpha \Delta \mathbf{u}^{(i)}$ 
31: end for
```

---

**Evaluation and reuse.** Full Newton evaluates tangents at all candidates, including rejected ones, and reuses the accepted assembly at the next iteration. Modified Newton evaluates a tangent only at iteration zero and reuses its KKT factorization; its candidates are residual-only. The merit scales stay fixed during each search. Recoverable candidate failures reject that length; search acceptance does not commit history. With backtracking disabled, omit the search and take  $\alpha = 1$  without a candidate evaluation or cache.

Recoverable attempt failure discards trial data, caches, and factors; the time controller cuts back and retries from the unchanged committed state with a fresh first tangent. Increment/retry limits may stop the analysis; model/setup errors are fatal. After acceptance, the controller advances time, handles output/restart events, and applies the growth rule.

Cold start initializes history and zero stress; restart restores committed history and accepted stress. Initial output uses this response without zero-time constitutive evolution. Optional final tangent verification uses the original positive-increment history before commitment.

## A Example constitutive plug-in: finite-strain J2 plasticity

This appendix defines one stateful material that can be supplied through the generic material-point interface. It is deliberately outside the main formulation: none of the finite element, constraint, solution, or output equations depends on J2 plasticity. Another material may replace this appendix's update provided it accepts the same kinematic inputs and returns the required stress, state, status, and endpoint tangent.

### A.1 Kinematics, elastic response, and history

Use the multiplicative split

$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_p, \quad \det \mathbf{F}_p = 1, \quad \mathbf{C}_p = \mathbf{F}_p^\top \mathbf{F}_p. \quad (155)$$

The committed history is the inverse plastic metric  $\mathbf{C}_{p,n}^{-1}$  and accumulated equivalent plastic strain  $\bar{\varepsilon}_{p,n}$ . The cold state is

$$\mathbf{C}_{p,0}^{-1} = \mathbf{I}, \quad \bar{\varepsilon}_{p,0} = 0. \quad (156)$$

No deformation gradient is stored as history.

For elastic Jacobian  $J_e = \det \mathbf{F}_e = J$  and unimodular elastic left metric

$$\bar{\mathbf{b}}_e = J^{-2/3} \mathbf{F} \mathbf{C}_p^{-1} \mathbf{F}^\top, \quad \det \bar{\mathbf{b}}_e = 1, \quad (157)$$

the stored energy selected for this material is

$$W_e = \frac{K}{2} (J - 1)^2 + \frac{\mu}{2} (\text{tr} \bar{\mathbf{b}}_e - 3). \quad (158)$$

Thus the Kirchhoff stress is

$$\boldsymbol{\tau} = KJ(J - 1)\mathbf{I} + \mathbf{s}, \quad \mathbf{s} = \mu \text{dev} \bar{\mathbf{b}}_e. \quad (159)$$

This volumetric energy is specific to this plug-in and is not the coupled logarithmic-volume neo-Hookean reference material used elsewhere.

The isotropic Voce law with a linear tail is

$$h(\bar{\varepsilon}_p) = \sigma_{y0} + H\bar{\varepsilon}_p + Q(1 - \exp[-b\bar{\varepsilon}_p]), \quad (160)$$

$$h'(\bar{\varepsilon}_p) = H + Qb \exp[-b\bar{\varepsilon}_p], \quad (161)$$

and the yield function is

$$\Phi = \|\mathbf{s}\| - \sqrt{\frac{2}{3}} h(\bar{\varepsilon}_p). \quad (162)$$

Here  $\|\mathbf{s}\| = \sqrt{\mathbf{s} : \mathbf{s}}$  is the spatial Frobenius norm.

### A.2 Elastic predictor and radial return

At every global Newton trial, hold the committed state fixed and form

$$\bar{\mathbf{b}}^{\text{tr}} = J_{n+1}^{-2/3} \mathbf{F}_{n+1} \mathbf{C}_{p,n}^{-1} \mathbf{F}_{n+1}^\top, \quad (163)$$

$$\mathbf{s}^{\text{tr}} = \mu \text{dev} \bar{\mathbf{b}}^{\text{tr}}, \quad \mathbf{q}^{\text{tr}} = \|\mathbf{s}^{\text{tr}}\|, \quad (164)$$

$$\bar{\mu} = \frac{\mu}{3} \text{tr} \bar{\mathbf{b}}^{\text{tr}}. \quad (165)$$



If  $q^{\text{tr}} - \sqrt{2/3} h(\bar{\varepsilon}_{\text{p},n})$  is not positive beyond the yield tolerance, accept the elastic predictor and leave history unchanged. Otherwise solve the monotone scalar equation

$$r(\Delta\gamma) = q^{\text{tr}} - 2\bar{\mu}\Delta\gamma - \sqrt{\frac{2}{3}} h\left(\bar{\varepsilon}_{\text{p},n} + \sqrt{\frac{2}{3}}\Delta\gamma\right) = 0. \quad (166)$$

Its derivative is

$$r'(\Delta\gamma) = -2\bar{\mu} - \frac{2}{3}h'\left(\bar{\varepsilon}_{\text{p},n} + \sqrt{\frac{2}{3}}\Delta\gamma\right) < 0. \quad (167)$$

The update is

$$\beta = 1 - \frac{2\bar{\mu}\Delta\gamma}{q^{\text{tr}}}, \quad \mathbf{s}_{n+1} = \beta \mathbf{s}^{\text{tr}}, \quad (168)$$

$$\bar{\varepsilon}_{\text{p},n+1} = \bar{\varepsilon}_{\text{p},n} + \sqrt{\frac{2}{3}}\Delta\gamma. \quad (169)$$

The radial stress correction determines only the deviatoric part of the updated unimodular elastic metric. Define  $\mathbf{q} = \mathbf{s}_{n+1}/\mu$  and choose the unique scalar  $c > -\lambda_{\min}(\mathbf{q})$  satisfying

$$\det(\mathbf{q} + c\mathbf{I}) = 1. \quad (170)$$

Then reconstruct

$$\bar{\mathbf{b}}_{\text{e},n+1} = \mathbf{q} + c\mathbf{I}, \quad \mathbf{C}_{\text{p},n+1}^{-1} = J_{n+1}^{2/3} \mathbf{F}_{n+1}^{-1} \bar{\mathbf{b}}_{\text{e},n+1} \mathbf{F}_{n+1}^{-\top}. \quad (171)$$

This documented spherical reconstruction leaves the radial stress correction unchanged while enforcing symmetry, positive definiteness, and  $\det \mathbf{C}_{\text{p},n+1}^{-1} = 1$  to numerical tolerance. It avoids an ad hoc normalization of a nonsymmetric or indefinite state after the fact.

Finally,

$$\mathbf{P}_{n+1} = \tau_{n+1} \mathbf{F}_{n+1}^{-\top}. \quad (172)$$

### A.3 Exact linearization of the discrete update

The implemented tangent differentiates the preceding discrete return, not the continuum rate equations with frozen internal variables. For an arbitrary direction  $\mathbf{H} = \delta\mathbf{F}$ , define

$$a = \text{tr}(\mathbf{F}^{-1}\mathbf{H}), \quad \delta J = Ja. \quad (173)$$

With committed  $\mathbf{C}_{\text{p},n}^{-1}$  fixed,

$$\delta \bar{\mathbf{b}}^{\text{tr}} = -\frac{2}{3}a\bar{\mathbf{b}}^{\text{tr}} + J^{-2/3} \left( \mathbf{H} \mathbf{C}_{\text{p},n}^{-1} \mathbf{F}^{\top} + \mathbf{F} \mathbf{C}_{\text{p},n}^{-1} \mathbf{H}^{\top} \right), \quad (174)$$

$$\delta \mathbf{s}^{\text{tr}} = \mu \text{dev}(\delta \bar{\mathbf{b}}^{\text{tr}}), \quad (175)$$

$$\delta q^{\text{tr}} = \frac{\mathbf{s}^{\text{tr}} : \delta \mathbf{s}^{\text{tr}}}{q^{\text{tr}}}, \quad \delta \bar{\mu} = \frac{\mu}{3} \text{tr}(\delta \bar{\mathbf{b}}^{\text{tr}}). \quad (176)$$

On a smooth plastic branch, implicit differentiation of Eq. (166) gives

$$\delta \Delta\gamma = \frac{\delta q^{\text{tr}} - 2\Delta\gamma \delta \bar{\mu}}{2\bar{\mu} + \frac{2}{3}h'(\bar{\varepsilon}_{\text{p},n+1})}. \quad (177)$$

Consequently,

$$\delta \beta = -2 \left[ \frac{\Delta\gamma \delta \bar{\mu} + \bar{\mu} \delta \Delta\gamma}{q^{\text{tr}}} - \frac{\bar{\mu} \Delta\gamma \delta q^{\text{tr}}}{(q^{\text{tr}})^2} \right], \quad (178)$$

$$\delta \mathbf{s}_{n+1} = \delta \beta \mathbf{s}^{\text{tr}} + \beta \delta \mathbf{s}^{\text{tr}}. \quad (179)$$

On the elastic branch, simply use  $\delta \mathbf{s}_{n+1} = \delta \mathbf{s}^{\text{tr}}$ . Both branches finish with

$$\delta \boldsymbol{\tau} = K(2J - 1)\delta J \mathbf{I} + \delta \mathbf{s}_{n+1}, \quad (180)$$

$$\delta \mathbf{P} = \delta \boldsymbol{\tau} \mathbf{F}^{-\top} - \boldsymbol{\tau} \mathbf{F}^{-\top} \mathbf{H}^\top \mathbf{F}^{-\top}. \quad (181)$$

Equation (181) is evaluated for the nine Cartesian basis directions to populate  $A_{iIjJ}^{\text{alg}}$ . This is an analytic directional construction of the exact algorithmic tangent; it is not finite differencing. At the elastic–plastic switching surface the update is nonsmooth and a unique classical tangent is not required. Centered finite differences away from that surface remain the independent verification oracle.

The trial state produced above is committed only after global convergence. A cutback repeats the update from the unchanged  $\mathbf{C}_{p,n}^{-1}$  and  $\bar{\varepsilon}_{p,n}$ .

## B Additional tangent remarks and crystal-plasticity differentiation

### B.1 Increment notation versus endpoint notation

A solver-specific constitutive Jacobian is often written symbolically as

$$\frac{\partial \Delta \text{stress}}{\partial \Delta \text{strain}}. \quad (182)$$

During a global Newton perturbation the beginning-of-increment state is fixed. Therefore

$$\delta(\Delta \text{stress}) = \delta(\text{stress}_{n+1}), \quad \delta(\Delta \text{strain}) = \delta(\text{strain}_{n+1}). \quad (183)$$

The word “increment” does not imply that history sensitivity is neglected. It specifies the discrete map whose endpoint is being differentiated.

### B.2 Numerical differentiation of a crystal-plasticity update

Suppose  $\mathbf{z}$  collects the local constitutive unknowns solved inside one material-point update, and the update is defined implicitly by

$$\mathbf{r}(\mathbf{z}_{n+1}, \mathbf{F}_{n+1}; \mathbf{z}_n, \mathbf{F}_n, \Delta t) = \mathbf{0}, \quad (184)$$

and  $\mathbf{P}_{n+1} = \hat{\mathbf{P}}(\mathbf{F}_{n+1}, \mathbf{z}_{n+1}; \mathbf{F}_n, \mathbf{z}_n, \Delta t)$ . The consistent algorithmic derivative is

$$\frac{D\mathbf{P}_{n+1}}{D\mathbf{F}_{n+1}} = \frac{\partial \hat{\mathbf{P}}}{\partial \mathbf{F}_{n+1}} + \frac{\partial \hat{\mathbf{P}}}{\partial \mathbf{z}_{n+1}} : \frac{D\mathbf{z}_{n+1}}{D\mathbf{F}_{n+1}}. \quad (185)$$

Thus, a finite-difference evaluation of  $D\mathbf{P}/D\mathbf{F}$  is consistent only if each perturbed  $\mathbf{F}_{n+1}$  is passed through the complete local constitutive solve from the same committed state at  $t_n$ . Holding the converged internal variables fixed would instead produce a frozen-state derivative.

For active-set plasticity or crystal plasticity, keeping the currently active slip set fixed while resolving the local return-mapping equations gives the tangent of that smooth active-set branch. At an activation or deactivation boundary the constitutive map is generally nonsmooth; a unique classical derivative need not exist. This is not an error in the finite element formulation. It is a property of the local constitutive map.

A perturbation magnitude such as  $10^{-8}$  may be suitable when components of  $\mathbf{F}$  are of order unity and double precision is used, but the perturbation should eventually be checked for truncation and cancellation sensitivity. A scale-aware perturbation and a central-difference verification are useful implementation tests.

## C Reference-form cross-check of the spatial linearization

The same internal virtual work can be pulled back to the fixed reference configuration,

$$G^{\text{int}} = \int_{\Omega_0} \mathbf{P} : \text{Grad}_X \mathbf{v} \, dV. \quad (186)$$

Because  $\text{Grad}_X \mathbf{v}$  and  $dV$  are fixed during the directional derivative with respect to the current nodal configuration,

$$DG^{\text{int}}[\Delta \mathbf{u}] = \int_{\Omega_0} \text{Grad}_X \mathbf{v} : \mathbb{A}^{\text{alg}} : \text{Grad}_X \Delta \mathbf{u} \, dV. \quad (187)$$

Using

$$\text{Grad}_X \mathbf{v} = \mathbf{G}_v \mathbf{F}, \quad \text{Grad}_X \Delta \mathbf{u} = \mathbf{h} \mathbf{F}, \quad dV = \frac{1}{J} dv, \quad (188)$$

Eq. (187) becomes

$$DG^{\text{int}}[\Delta \mathbf{u}] = \int_{\Omega_t} \frac{1}{J} (\mathbf{G}_v \mathbf{F}) : \mathbb{A}^{\text{alg}} : (\mathbf{h} \mathbf{F}) \, dv. \quad (189)$$

Substitution of the push-forward relation of Section 6 reduces Eq. (189) exactly to Eq. (51). This identity is a useful independent audit of the spatial derivation. It also clarifies the status of the geometric stiffness: when the endpoint constitutive derivative is kept in the material pair  $(\mathbf{P}, \mathbf{F})$  on  $\Omega_0$ , the same total directional derivative is represented by the single  $\mathbb{A}^{\text{alg}}$  term. After rewriting that derivative in current-configuration variables, it separates naturally into a spatial constitutive contribution and the conventional initial-stress/geometric contribution.

## D Near-incompressibility and element-technology alternatives

Plastic incompressibility does not by itself imply total incompressibility. In a multiplicative crystal-plasticity split

$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_p, \quad (190)$$

isochoric plastic flow gives

$$\det \mathbf{F}_p = 1, \quad (191)$$

and therefore

$$J = \det \mathbf{F} = \det \mathbf{F}_e. \quad (192)$$

A material may therefore have isochoric plastic flow while retaining finite elastic compressibility. Nevertheless, once plastic flow dominates the incremental response, a displacement formulation can behave numerically as a nearly incompressible problem. Fully integrated displacement elements can develop volumetric locking as the constitutive response approaches near-incompressibility; this must be assessed for the intended elastic-plastic parameter range.

The unstabilized reference formulations in this document are standard displacement-only Hex8 with full  $2^3$  integration and standard displacement-only Hex20 with full  $3^3$  integration. The selected anti-locking option is the separate centroidal  $\bar{F}$  Hex8 formulation. This keeps the constitutive tangent representation independent of stabilization for the two standard elements; locking must therefore be tested rather than assumed absent.

### D.1 How the main remedies alter the present formulation

| Method                             | Change relative to the present element                                                                                                                                  | New global unknowns                               | Main consequence for this formulation                                                                                                                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uniform reduced integration        | Replace $3^3$ quadrature by a lower-order rule; constitutive map and weak form are otherwise unchanged.                                                                 | None                                              | Smallest code and derivation change, but under-integration can introduce zero-energy modes. For Hex20 it is a small formal change, but under-integration can still introduce zero-energy modes and is therefore not selected in the present implementation scope. |
| Selective reduced integration      | Integrate volumetric and deviatoric contributions with different rules.                                                                                                 | None                                              | Requires a defensible volumetric/deviatoric split of the residual and tangent. This is less neutral for a generic anisotropic crystal-plasticity material interface.                                                                                              |
| $\bar{B}$ / projection             | Replace the compatible dilatational strain or virtual-strain operator by a projected/averaged field.                                                                    | None                                              | Keeps a displacement-only global system, but changes the strain-displacement operator and its consistent linearization. The finite-strain extension is more involved than changing quadrature.                                                                    |
| $\bar{F}$ / projection             | Replace the compatible $\mathbf{F}$ supplied to the constitutive response by a modified deformation gradient whose volumetric part is projected to a lower-order space. | None                                              | Particularly natural for the present $\mathbf{P}(\mathbf{F})$ and $D\mathbf{P}/D\mathbf{F}$ material interface. Stress, internal force, and tangent must all use the modified kinematics consistently.                                                            |
| Mixed $u$ - $p$ or $u$ - $p$ - $J$ | Add independently interpolated pressure and possibly volume/Jacobian fields and their weak equations.                                                                   | Yes, unless local fields are statically condensed | Largest change: new variational equations, interpolation/stability choices, block element matrices, assembly, constraints, and linear-solver considerations. It is the most direct path toward the truly incompressible limit.                                    |

For the present project, the classical centroidal  $\bar{F}$  Hex8 of Section 15 is now the selected displacement-only anti-locking element. It preserves full  $2 \times 2 \times 2$  integration, introduces no new global unknowns, and preserves the native material interface based on two deformation gradients and  $D\mathbf{P}/D\mathbf{F}$ . Standard Hex8 remains available as the unstabilized control element.

A mixed formulation remains the cleaner choice if the intended material approaches exact incompressibility, if pressure itself is an important solution quantity, or if a projection formulation proves insufficiently robust. It should be regarded as a new element family rather than a small option inside the present element kernel.

## D.2 Published and public implementation references

Several public implementations are useful as design references without being copied directly. MOOSE implements finite-deformation  $\bar{F}$  stabilization in both total- and updated-Lagrangian kernels and emphasizes that changing the deformation gradient also changes the consistent Jacobian. OpenGeoSys documents and implements an  $\bar{F}$  large-deformation process with both element-center and element-average volumetric measures and publishes the corresponding linearization. The deal.II step-44 tutorial provides a complete three-field finite-strain quasi-incompressible solver with displacement, pressure, and independent volume/Jacobian fields. These implementations illustrate the relative software impact of projection and mixed approaches.

The standard and  $\bar{F}$  Hex8 elements should be compared over the intended elastic bulk modulus, plastic strain range, mesh refinement, and element distortion before the projection is trusted for crystal plasticity. Pressure oscillation/checkerboarding, excessive stiffness, and convergence behavior should be monitored together rather than judging locking from displacement alone. Standard Hex20 provides an additional higher-order reference but is not assumed to be locking-free.

## E Hexahedral shape functions and local-node conventions

### E.1 Hex8

Let the eight corner nodes have parent-coordinate signs

$$(\xi_A, \eta_A, \zeta_A) \in \{-1, +1\}^3. \quad (193)$$

The tri-linear Hex8 shape functions are

$$N_A(\xi, \eta, \zeta) = \frac{1}{8}(1 + \xi\xi_A)(1 + \eta\eta_A)(1 + \zeta\zeta_A). \quad (194)$$

Their parent derivatives are

$$N_{A,\xi} = \frac{1}{8}\xi_A(1 + \eta\eta_A)(1 + \zeta\zeta_A), \quad (195)$$

$$N_{A,\eta} = \frac{1}{8}\eta_A(1 + \xi\xi_A)(1 + \zeta\zeta_A), \quad (196)$$

$$N_{A,\zeta} = \frac{1}{8}\zeta_A(1 + \xi\xi_A)(1 + \eta\eta_A). \quad (197)$$

For the centroidal  $\bar{F}$  element these same shape functions are evaluated at the eight Gauss points and once at  $\boldsymbol{\xi}_c = (0, 0, 0)^\top$  for the projection kinematics. No separate centroid shape functions or internal degrees of freedom are introduced.

### E.2 Hex20

Let corner-node signs be

$$(\xi_A, \eta_A, \zeta_A) \in \{-1, +1\}^3. \quad (198)$$

For each of the eight corner nodes,

$$N_A(\xi, \eta, \zeta) = \frac{1}{8}(1 + \xi\xi_A)(1 + \eta\eta_A)(1 + \zeta\zeta_A)(\xi\xi_A + \eta\eta_A + \zeta\zeta_A - 2). \quad (199)$$

For an edge midpoint on an edge parallel to the  $\xi$  direction, with fixed  $\eta_A, \zeta_A \in \{-1, +1\}$ ,

$$N_A = \frac{1}{4}(1 - \xi^2)(1 + \eta\eta_A)(1 + \zeta\zeta_A). \quad (200)$$

For an edge parallel to  $\eta$ ,

$$N_A = \frac{1}{4}(1 - \eta^2)(1 + \xi\xi_A)(1 + \zeta\zeta_A). \quad (201)$$

For an edge parallel to  $\zeta$ ,

$$N_A = \frac{1}{4}(1 - \zeta^2)(1 + \xi\xi_A)(1 + \eta\eta_A). \quad (202)$$

These expressions define all 20 serendipity shape functions. Their parent-coordinate derivatives are obtained analytically from Eqs. (199)–(202).

A concrete node numbering is not part of the continuum formulation, but any implementation must use one ordering consistently in interpolation, connectivity, face definitions, and constraint generation.

## F Guide to variations and consistent linearization

This appendix is a reading guide for the variational steps used in Part II. Its purpose is not to introduce a second formulation, but to make explicit which quantities depend on the nonlinear displacement field and which are held fixed during a directional derivative.

### F.1 Variation as a directional derivative

For a functional  $Q(\mathbf{u})$ , the variation in direction  $\Delta\mathbf{u}$  is

$$\delta Q \equiv DQ(\mathbf{u})[\Delta\mathbf{u}] = \left. \frac{d}{d\epsilon} Q(\mathbf{u} + \epsilon\Delta\mathbf{u}) \right|_{\epsilon=0}. \quad (203)$$

The operator is linear and obeys the ordinary product and chain rules. What matters is the dependency graph. A factor that does not depend on  $\mathbf{u}$  has zero variation.

### F.2 Product rule and dependency bookkeeping

Consider a product  $A(\mathbf{u})B(\mathbf{u})$ . If both factors depend on the perturbed displacement field, then

$$\delta(AB) = (\delta A)B + A(\delta B). \quad (204)$$

If instead  $B = B_0$  is fixed with respect to the chosen perturbation,

$$\delta(A(\mathbf{u})B_0) = (\delta A)B_0, \quad \delta B_0 = 0. \quad (205)$$

Nothing has happened to the Leibniz rule; one of its two terms is simply zero. The main task in a nonlinear FE linearization is therefore to identify the dependency of every factor before applying the variation.

For example, in the reference-form internal work

$$G^{\text{int}} = \int_{\Omega_0} \mathbf{P} : \text{Grad}_X \mathbf{v} \, dV, \quad (206)$$

$\mathbf{P}$  depends on the current trial endpoint, whereas  $\text{Grad}_X \mathbf{v}$  and  $dV$  are fixed because they are referred to  $\Omega_0$  and the test coefficients are held fixed. Hence only  $\mathbf{P}$  contributes a nonzero variation. In the spatial form, by contrast,  $\boldsymbol{\sigma}$ ,  $\nabla_x \mathbf{v}$ , and  $dv$  all depend on the current map and all three product-rule contributions must initially be retained; some later cancel after the Truesdell relation is substituted.

### F.3 Test field versus Newton direction

Let  $\mathcal{V}_0$  denote the admissible test space satisfying the homogeneous form of the essential displacement conditions. The weak residual is tested with  $\mathbf{v} \in \mathcal{V}_0$ ,

$$G(\mathbf{u}, \mathbf{v}) = 0 \quad \forall \mathbf{v} \in \mathcal{V}_0. \quad (207)$$

while Newton requires

$$DG(\mathbf{u}, \mathbf{v})[\Delta \mathbf{u}]. \quad (208)$$

During this derivative,  $\Delta \mathbf{u}$  is the perturbation direction and the nodal coefficients of  $\mathbf{v}$  are fixed. Thus there is no term representing a variation of those coefficients. Nevertheless, spatial derivatives of  $\mathbf{v}$  can vary because the operator  $\nabla_x$  depends on the current map.

### F.4 Basic kinematic identities

Let

$$\mathbf{h} = \delta \mathbf{F} \mathbf{F}^{-1}. \quad (209)$$

Then

$$\delta J = J \operatorname{tr}(\mathbf{h}), \quad (210)$$

$$\delta(\mathbf{F}^{-1}) = -\mathbf{F}^{-1}(\delta \mathbf{F})\mathbf{F}^{-1}, \quad (211)$$

$$\delta(\mathbf{F}^{-\top}) = -\mathbf{F}^{-\top}(\delta \mathbf{F})^\top \mathbf{F}^{-\top}. \quad (212)$$

These are direct applications of the chain rule to  $J(\mathbf{F})$  and  $\mathbf{F}^{-1}\mathbf{F} = \mathbf{I}$ .

### F.5 Why the spatial test-function gradient varies

Using  $\nabla_x \mathbf{v} = (\operatorname{Grad}_X \mathbf{v})\mathbf{F}^{-1}$  and holding  $\operatorname{Grad}_X \mathbf{v}$  fixed,

$$\delta(\nabla_x \mathbf{v}) = (\operatorname{Grad}_X \mathbf{v}) \delta(\mathbf{F}^{-1}) \quad (213)$$

$$= -(\operatorname{Grad}_X \mathbf{v})\mathbf{F}^{-1}(\delta \mathbf{F})\mathbf{F}^{-1} \quad (214)$$

$$= -(\nabla_x \mathbf{v})\mathbf{h}. \quad (215)$$

Only the inverse deformation gradient is varied because the reference-gradient representation of the fixed test field does not depend on the current trial displacement.

### F.6 Variation of the current volume element

Because  $dv = J dV$  and  $dV$  belongs to the fixed reference configuration,

$$\delta(dv) = \delta J dV = \operatorname{tr}(\mathbf{h}) dv. \quad (216)$$

Again, only  $J$  varies;  $dV$  does not.

### F.7 Example: internal virtual work

For

$$G^{\text{int}} = \int_{\Omega_t} \boldsymbol{\sigma} : \nabla_x \mathbf{v} dv, \quad (217)$$

all three displayed factors have different dependency status during Newton linearization:  $\boldsymbol{\sigma}$  depends on the trial kinematics through the constitutive update;  $\nabla_x \mathbf{v}$  depends on the current map even though the test coefficients are fixed; and  $d\mathbf{v}$  depends on  $J$ . Therefore

$$\delta G^{\text{int}} = \int_{\Omega_t} [\delta \boldsymbol{\sigma} : \nabla_x \mathbf{v} + \boldsymbol{\sigma} : \delta(\nabla_x \mathbf{v}) + \boldsymbol{\sigma} : \nabla_x \mathbf{v} \operatorname{tr}(\mathbf{h})] d\mathbf{v}. \quad (218)$$

Substitution of the preceding identities gives the expression used in Section 9.

## F.8 Constitutive variables are varied implicitly through the algorithmic tangent

For a history-dependent material, the endpoint stress has the condensed form

$$\mathbf{P}_{n+1} = \hat{\mathbf{P}}(\mathbf{F}_{n+1}; \mathbf{F}_n, \mathcal{H}_n, \Delta t), \quad (219)$$

where the local integration algorithm has already solved for trial internal variables. The global FE linearization therefore uses

$$\delta \mathbf{P}_{n+1} = \mathbb{A}^{\text{alg}} : \delta \mathbf{F}_{n+1}. \quad (220)$$

One does not add a second explicit  $\partial \mathbf{P} / \partial \mathcal{H} \delta \mathcal{H}$  term at global level if  $\mathbb{A}^{\text{alg}}$  is the derivative of the complete condensed local update; that dependency is already contained in  $\mathbb{A}^{\text{alg}}$ .

## F.9 Reference-form linearization as a useful check

The pull-back

$$G^{\text{int}} = \int_{\Omega_0} \mathbf{P} : \operatorname{Grad}_X \mathbf{v} dV \quad (221)$$

is especially transparent:  $\operatorname{Grad}_X \mathbf{v}$  and  $dV$  are fixed, so

$$\delta G^{\text{int}} = \int_{\Omega_0} \operatorname{Grad}_X \mathbf{v} : \mathbb{A}^{\text{alg}} : \operatorname{Grad}_X \Delta \mathbf{u} dV. \quad (222)$$

When this expression is pushed forward to the current configuration, the same material and geometric tangent terms obtained in Part II must reappear. This equivalence is a strong algebraic and implementation check.

## G References

The principal formulation/notation reference is T. Belytschko, W. K. Liu, B. Moran, and K. Elkhodary, *Nonlinear Finite Elements for Continua and Structures*, 2nd ed., Wiley, 2014.

The finite-strain J2 example in Appendix A follows the multiplicative hyperelastic-plastic framework and radial-return structure of J. C. Simo, “Algorithms for static and dynamic multiplicative plasticity that preserve the classical return mapping schemes of the infinitesimal theory,” *Computer Methods in Applied Mechanics and Engineering*, 99, 61–112 (1992). Its determinant-one spherical reconstruction is stated explicitly in Eqs. (170)–(171) as part of this solver’s discrete material definition.

For volume-constraint projection and nearly incompressible finite deformation, useful references include J. C. Simo, R. L. Taylor, and K. S. Pister, “Variational and projection methods for the volume constraint in finite deformation elasto-plasticity,” *Computer Methods in Applied Mechanics and Engineering*, 51, 177–208 (1985); E. A. de Souza Neto, D. Perić, M. Dutko, and D. R. J. Owen, “Design of simple low order finite elements for large strain analysis of nearly incompressible solids,” *International Journal of Solids and Structures*, 33, 3277–3296 (1996); and T. Elguedj, Y. Bazilevs, V. M. Calo, and T. J. R. Hughes, “ $\bar{B}$  and  $\bar{F}$  projection methods for nearly incompressible linear and nonlinear elasticity and plasticity using higher-order NURBS elements,” *Computer Methods in Applied Mechanics and Engineering*, 197, 2732–2762 (2008).